PHYSICS

1) A charged particle is kept in equilibrium in an electric field between the plates in the Millikan oil drop experiment. If the direction of the electric field between the plates is reversed, then the acceleration of the charged particle will be :

(1) g

(2) 2g

(3) $\frac{g}{2}$

(4) $\frac{g}{4}$

2) A particle of mass m, charge –Q is constrained to move along the axis of a ring of radius a. The ring carries a uniform charge density + λ along its circumference. Initially, the particle lies in the plane of the ring at a point where no net force acts on it. The period of oscillation of the particle when it is displaced slightly from its equilibrium position is -

(1) $T = 4\pi\sqrt{\frac{\varepsilon_0 ma^2}{\lambda Q}}$

(2) $T = 2\pi\sqrt{\frac{2\varepsilon_0 ma^2}{\lambda Q}}$

(3) $T = 2\pi\sqrt{\frac{4\varepsilon_0 ma^2}{\lambda Q}}$

(4) $T = 2\pi\sqrt{\frac{\varepsilon_0 ma^2}{2\lambda Q}}$

3) What equal charges should to be placed on earth and moon to neutralize their gravitational attraction ?
(mass of earth = 10^{25} kg, mass of moon = 10^{23} kg)

(1) 8.6×10^{13} C

(2) 6.8×10^{26} C

(3) 8.6×10^{3} C

(4) 9×10^{6} C

4) Five positive equal charges are placed at vertices of a regular hexagon and net electric field at the centre is E_1. A negative charge having equal magnitude is placed sixth vertex and then net electric field is E_2. Find $\frac{E_2}{E_1}$.

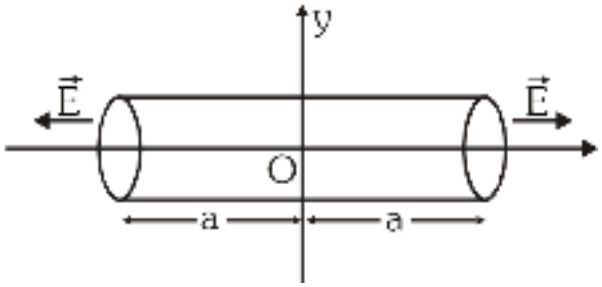
(1) 3

(2) 2

(3) 4

(4) 1

5) A right circular cylinder of length 2a cm and radius r cm has its centre at the origin O and its axis along the x-axis so that one flat face is at
x = +a cm and the other is at x = – a cm as is shown in the figure. It is placed in a uniform electric



field $\vec{E} = E_x\hat{i}$ NC^{-1} for x > 0 and $\vec{E} = -E_x\hat{i}$ for x < 0. Then

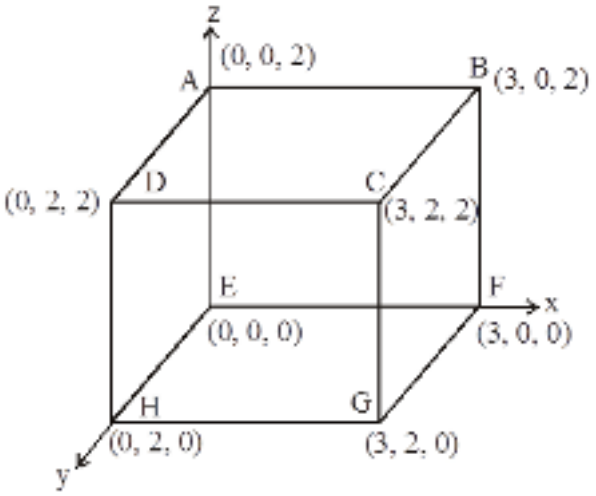
(1) The net outward flux through each flat surface is $E_a pr^2 \times 10^{-6}$ Nm^2/C

(2) The net outward flux through curved surface of the cylinder is non zero

(3) The net charge inside the cylinder is $E_a r^2 \times 5.5 \times 10^{-15}$ C

(4) The net charge inside the cylinder is $E_a r^2 \times 25 \times 10^{-15}$ C

6) An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j}\,N/C$ passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_I and ϕ_{II} respectively. The difference



between (ϕ_I – ϕ_{II}) is (in Nm^2/C) _____.

(1) – 48

(2) 48

(3) – 36

(4) 36

7) Two point charge of 100 μC and -4 μC are positioned at points (-2√3, 3√3, -4) and (4√3, -5√3, 6) respectively of a Cartesian coordinate system. Find the force vector on the -4 μC charge? All the coordinates are in meters.

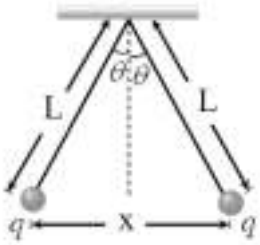
(1) $9\times10^{-4}\left(3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k}\right)$

(2) $9\times10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$

(3) $2.25\times10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$

(4) $2.25\times10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$

8) In the given figure two tiny conducting balls of identical mass m and identical charge q hang from non-conducting threads of equal length L. Assume that θ is so small that $\tan\theta \approx \sin\theta$ is equal to,



find the value of x.

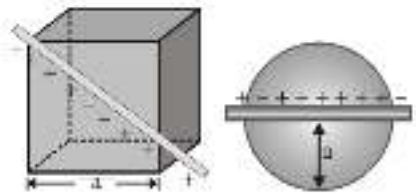
(1) $\left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(2) $\left(\frac{qL^2}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(3) $\left(\frac{q^2L^2}{4\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(4) $\left(\frac{q^2L}{4\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

9) A linear charge having linear charge density λ, penetrates a cube diagonally and then it penetrate a sphere diametrically as shown. What will be the ratio of flux coming cut of cube and sphere :-



(1) $\sqrt{3}:2$

(2) $2:\sqrt{3}$

(3) $\sqrt{3}:1$

(4) $\sqrt{3}:4$

10) Two positive charges of 20 coulomb and Q coulomb are situated at a distance of 60 cm. The neutral point between them is at a distance of 20 cm from the 20 coulomb charge. Charge Q (in coulomb) is:-

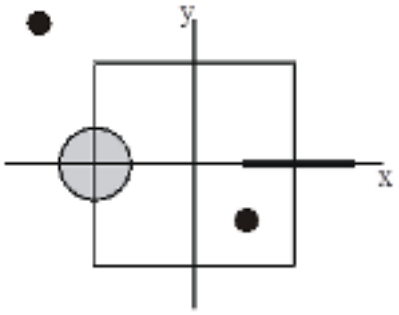
(1) 40

(2) 60

(3) 80

(4) 20

11) A disk of radius a/4 having a uniformly distributed charge 6C is placed in the x-y plane with its centre at (–a/2, 0, 0). A rod of length a carrying a uniformly distributed charge 8 C is placed on the x-axis from x = a/4 to x = 5a/4. Two point charges –7 C and 3 C are placed at (a/4, –a/4, 0) and (–3a/4, 3a/4, 0), respectively. Consider a cubical surface formed by six surfaces x = ± a/2, y = ± a/2, z = ± a/2. The electric flux through this cubical surface is :-



(1) $\frac{-2C}{\varepsilon_0}$

(2) $\frac{2C}{\varepsilon_0}$

(3) $\frac{10C}{\varepsilon_0}$

(4) $\frac{12C}{\varepsilon_0}$

12) A sphere of radius R carries volume charge density proportional to the square of the distance from the center : $\rho = Ar^2$, where A is a positive constant. At a distance of R/2 from the center, the magnitude of the electric field is :-

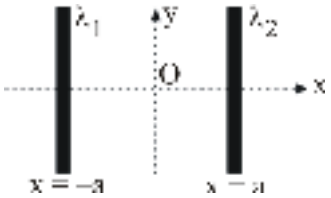
(1) $A/4\pi\varepsilon_0$

(2) $AR^3/40\varepsilon_0$

(3) $AR^3/24\varepsilon_0$

(4) $AR^3/5\varepsilon_0$

13) Two positively charged infinitely long wires ($\lambda_1 > \lambda_2$) are placed parallel to each other at x = –a and x= +a as in figure. find the x-coordinate of a point at which electric field is zero.



(1) $\frac{\lambda_1}{\lambda_2}a$

(2) $\frac{\lambda_2}{\lambda_1}a$

(3) $\frac{(\lambda_1 - \lambda_2)\,a}{\lambda_1 + \lambda_2}$

(4) $\left(\frac{\lambda_1 + \lambda_2}{\lambda_1 - \lambda_2}\right) a$

14) Two point charges q and –q are at positions(0, 0, d) and (0, 0, –d) respectively. What is the electric field at (a, 0, 0)?

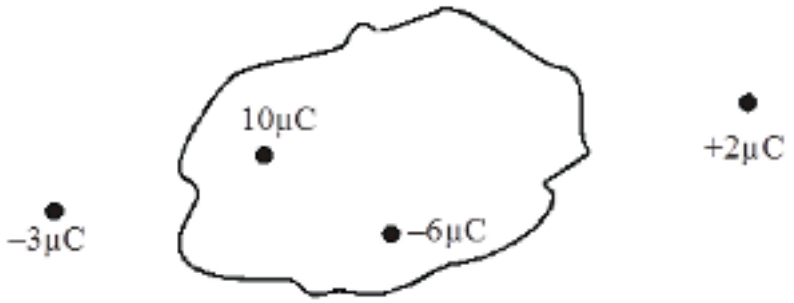
(1) $\frac{2qd}{4\pi\varepsilon_0(d^2 + a^2)^{3/2}}\hat{k}$

(2) $\frac{qd}{4\pi\varepsilon_0(d^2 + a^2)^{3/2}}\hat{k}$

(3) $\frac{-2qd}{4\pi\varepsilon_0(d^2 + a^2)^{3/2}}\hat{k}$

(4) $\frac{-qd}{4\pi\varepsilon_0(d^2 + a^2)^{3/2}}\hat{k}$

15) In a particular system, if number of electric field lines associated by 1C charge is 10^9. Then net number of electric field lines passing through the given closed surface is n × 10^3 find n.



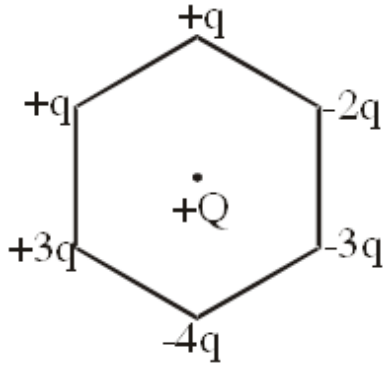
(1) 2

(2) 3

(3) 4

(4) 1

16) Six charges are kept at the vertices of a regular hexagon as shown in the figure. If magnitude of force applied by +Q on +q charge is F, then net electric force on the +Q is nF. Find the value of n.



(1) 6

(2) 5

(3) 4

(4) 9

17) Two infinitely long parallel wires having linear charge densities λ_1 and λ_2 respectively are placed at a distance of R. The force per unit length of an either wire will be $\left(k = \frac{1}{4\pi\epsilon_0}\right)$

(1) $\frac{k2\lambda_1\lambda_2}{R^2}$

(2) $\frac{k2\,\lambda_1\lambda_2}{R}$

(3) $\frac{k\lambda_1\lambda_2}{R^2}$

(4) $\frac{\lambda_1\lambda_2}{R}$

18) A particle moves with a velocity $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k})$ m/s under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})N.$ The instantaneous power applied to the particle is :-

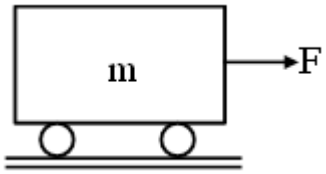
(1) 200 J/s

(2) 40 J/s

(3) 140 J/s

(4) 170 J/s

19) A car of mass m is accelerating on a level smooth road under the action of a single force F. The power delivered to the car is constant and equal to P. If the velocity of the car at an instant is v, then after travelling how much distance it becomes double ?



(1) $\frac{7mv^3}{3P}$

(2) $\frac{4mv^3}{3P}$

(3) $\frac{mv^3}{P}$

(4) $\frac{18mv^3}{7P}$

20)

A elevator that can carry a maximum load of 1500 kg (elevator + passenger) is moving up with a constant speed of 2m/s. The frictional force opposing the motion is 3000 N. Find the minimum power delivered by the motor to the elevator in watts.

(1) 15000 W

(2) 36000 W

(3) 72000 W

(4) 18000 W

21) An engine pumps up 100 kg of water through a height of 10 m in 5 s. Given that, the efficiency of engine is 60%. If g = 10 ms^{-2}, the power of this engine is:

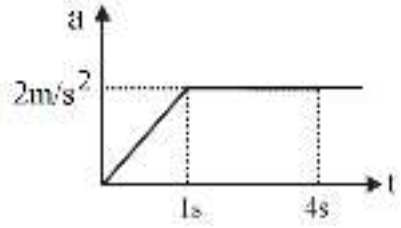
(1) 3.3 kW

(2) 0.33 kW

(3) 0.033 kW

(4) 33 kW

22) The acceleration time graph of a block of mass 4 kg moving along x-axis is as shown in the figure initial velocity of block is 2 m/s then the power delivered to block at t = 4s will be :



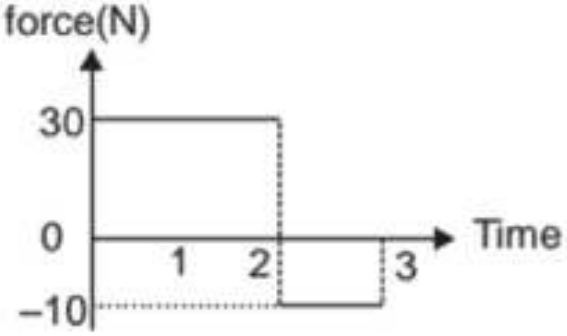
(1) 72 W

(2) 36 W

(3) 49 W

(4) 16 W

23) Starting at rest, a 10 kg object is acted upon by only one force as indicated in figure. Then the total work done by the force is.



(1) 90J

(2) 125J

(3) 245J

(4) 490J

24) A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table

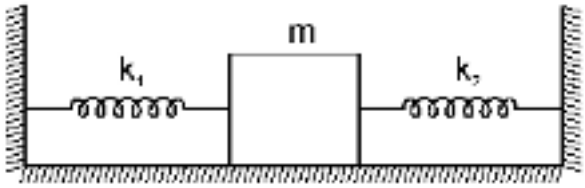
(1) 7.2 J

(2) 3.6 J

(3) 120 J

(4) 1200 J

25) A block of mass m is attached to two unstretched springs of spring constants k_1 and k_2 as shown in figure. The block is displaced towards right through a distance x and is released. Find the speed of the block as it passes through the natural length position shown.



(1) $\sqrt{\frac{k_1+k_2}{m}}\,x$

(2) $\sqrt{\frac{k_1k_2}{m(k_1+k_2)}}x$

(3) $\sqrt{\frac{k_1^2k_2^2}{m(k_1^2+k_2^2)}}x$

(4) $\sqrt{\frac{k_1^3k_2^3}{m(k_1^3+k_2^3)}}$

26) A particle is released from a height H. At certain height its kinetic energy is two times its potential energy. Height and speed of particle at that instant are :-

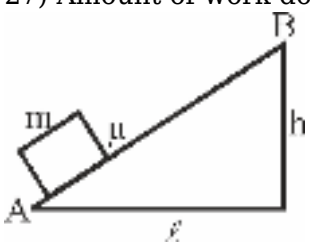
(1) $\frac{H}{3}, \sqrt{\frac{2gH}{3}}$

(2) $\frac{H}{3}, 2\sqrt{\frac{gH}{3}}$

(3) $\frac{2H}{3}, \sqrt{\frac{2gH}{3}}$

(4) $\frac{H}{3}, \sqrt{2gH}$

27) Amount of work done to carry a block from A to B will be (Assume friction coefficient μ)



(1) mgh

(2) $\mu mg\sqrt{l^2+h^2}$

(3) $\mu mg(\ell+h)$

(4) $mg(h + \mu\ell)$

28) A particle is released from rest at origin. It moves under influence of potential field $U = x^2 - 3x$, kinetic energy at x = 2 is.

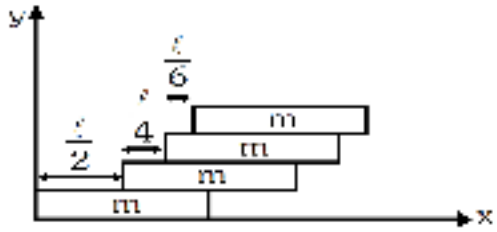
(1) 2J

(2) 1J

(3) 1.5J

(4) 0J

29) Find the x coordinate of the centre of mass of the bricks of length shown in figure :



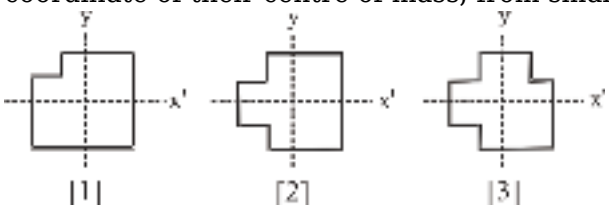
(1) $\frac{24}{25}\ell$

(2) $\frac{25}{24}\ell$

(3) $\frac{15}{16}\ell$

(4) $\frac{16}{15}\ell$

30) A machinist starts with three identical square plates but cuts one corner from one of them, two corners from the second and three corners from the third. Rank the three according to the x-coordinate of their centre of mass, from smallest to largest :-



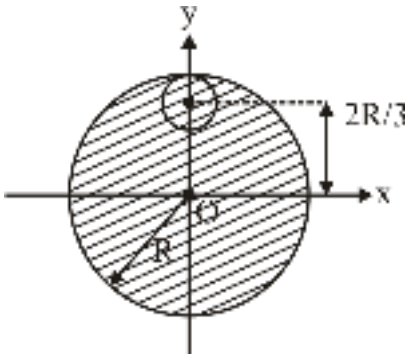
(1) 3,1,2

(2) 1,3,2

(3) 3,2,1

(4) 1 and 3 tie, then 2

31) From a circular disc of radius R and mass 9m, a small disc of radius R/3 is removed as shown in figure. The position of centre of mass of the remaining disc from centre of the original disc will be :-



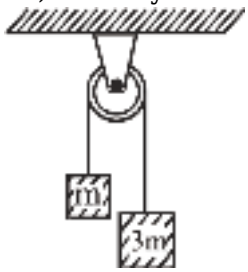
(1) $\left(0, -\frac{R}{12}\right)$

(2) $\left(-\frac{R}{12}, 0\right)$

(3) $\left(0, -\frac{R}{4}\right)$

(4) $\left(0, -\frac{R}{6}\right)$

32) If the system is released, then the acceleration of the centre of mass of the system is :-



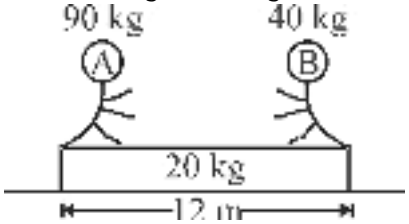
(1) g/4

(2) g/2

(3) g

(4) 2g

33) In the given diagram the distance travelled by the plank when A and B exchange their positions:



(1) 2 m

(2) 4 m

(3) 4.8 m

(4) 5.4 m

34) A cannon ball is fired with a velocity 200 m/sec at an angle of 60° with the horizontal. At the highest point of its flight it explodes into 3 equal fragments, one going vertically upwards with a velocity 100 m/sec, the second one falling vertically downwards with a velocity 100 m/sec. The third

fragment will be moving with a velocity

(1) 100 m/s in the horizontal direction

(2) 300 m/s in the horizontal direction

(3) 300 m/s in a direction making an angle of 60° with the horizontal

(4) 200 m/s in a direction making an angle of 60° with the horizontal

35) A particle of mass 5 kg initially at rest explodes into three fragments with mass ratio 3 : 1 : 1. Two of the fragments each of mass m are found to move with a speed 60 m/s in mutually perpendicular directions. The velocity of third fragment is:

(1) $60\sqrt{2}$ ms^{-1}

(2) $20\sqrt{3}$ ms^{-1}

(3) $10\sqrt{2}$ ms^{-1}

(4) $20\sqrt{2}$ ms^{-1}

36) A 10 g bullet, moving with a velocity of 500 m/s. enters a stationary piece of ice of mass 10 kg and stops. If the piece of ice is lying on a frictionless plane, then its velocity will be

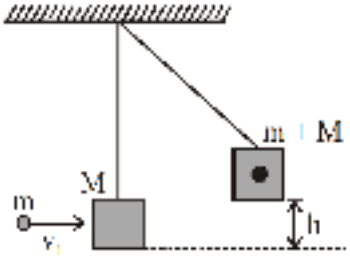
(1) 5 cm/s

(2) 5 m/s

(3) 0·5 m/s

(4) 0·5 cm/s

37) A bullet of mass m moving with a velocity v_1 strikes a suspended wooden block of mass M as shown in the figure and sticks to it. If the block rises to a height h then the initial velocity of the bullet is :-



(1) $\frac{m+M}{m}\sqrt{2gh}$

(2) $\sqrt{2gh}$

(3) $\frac{M+m}{M}\sqrt{2gh}$

(4) $\frac{m}{M+m}\sqrt{2gh}$

38) A particle moving along x-axis has acceleration f, at time t, given by $f = f_0\left(1-\frac{t}{T}\right)$, where f^0 and T are constants. The particle at t = 0 has zero velocity. At the instant when f = 0, the particle's velocity (v) is :

(1) f_0T

(2) $\frac{1}{2}f_0T^2$

(3) f_0T^2

(4) $\frac{1}{2}f_0T$

39) A stone is dropped from the top of a tower of height H. After 2s another stone is dropped from the balcony 60 m below the top. Both reach the bottom simultaneously. What is the value of H:-

(1) 45 m

(2) 80 m

(3) 125 m

(4) 180 m

40) A body thrown vertically upwards with an initial velocity u reaches maximum height in 6s. The ratio of the distances traveled by the body in the first second and the seventh second is :-

(1) 1 : 1

(2) 11 : 1

(3) 1 : 2

(4) 1 : 11

41) The distance between two particles moving towards each other is decreasing at the rate of 6 m/sec. If these particles travel with same speeds and in the same direction, then the separation increase at the rate of 4 m/sec. The particles have speeds as

(1) 5 m/sec ; 1 m/sec

(2) 4 m/sec ; 1 m/sec

(3) 4 m/sec ; 2 m/sec

(4) 5 m/sec ; 2 m/sec

42)

Imagine yourself standing in an elevator which is moving with an upward acceleration a = 2 ms^{-2}. A coin is dropped from rest from the roof of the elevator, relative to you. The roof to floor height of elevator is 1.5 m (take g = 10 ms^{-2}). Find time taken by the coin to reach at the floor of lift :-

(1) 1 sec

(2) $\frac{1}{2}$ sec

(3) 2 sec

(4) $\frac{\sqrt{3}}{2}$ sec

43) A swimmer can cross a 50 m wide river in a minimum time of 10 minute. He wishes to cross the river through shortest path but realises that river flows with a speed of 3 m/min. Find the angle with direction of river flow with which he should swim in order to fulfill his wish :-

(1) 37°

(2) 127°

(3) 53°

(4) 143°

44) The position of a particle moving in the xy-plane at any time t is given by $x = (3t^2 - 6t)$ metres, $y = (t^2 - 2t)$ metres. Select the correct statement about the moving particle from the following :-

(1) The acceleration of the particle is zero at t = 0 second

(2) The velocity of the particle is zero at t = 0 second

(3) The velocity of the particle is zero at t = 1 second

(4) The velocity and acceleration of the particle are never zero

45) The co-ordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time t is given by :

(1) $3t\sqrt{\alpha^2 + \beta^2}$

(2) $3t^2\sqrt{\alpha^2 + \beta^2}$

(3) $t^2\sqrt{\alpha^2 + \beta^2}$

(4) $\sqrt{\alpha^2 + \beta^2}$

CHEMISTRY

1) Which of the following have one lone pair on the central atom ?
(I) NH_3
(II) SF_4
(III) XeO_4
(IV) XeF_6

(1) I, II

(2) I, II, IV

(3) I, IV

(4) I, II,III, IV

2) The correct order of lattice energy of the following compounds

(1) $NaCl > MgBr_2 > CaO > Al_2O_3$

(2) $Al_2O_3 > MgBr_2 > CaO > NaCl$

(3) $MgBr_2 > Al_2O_3 > CaO > NaCl$

(4) $Al_2O_3 > CaO > MgBr_2 > NaCl$

3) Select incorrect order :-

(1) $Li^+(aq) > Na^+(aq) > K^+(aq) > Cs^+(aq)$ Ionic mobility order

(2) $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$ Hydrate size of ion

(3) $F^-(aq) > Cl^-(aq) > Br^-(aq) > I^-$ Hydrate size of ion

(4) $F^-(aq) < Cl^-(aq) < Br^-(aq) < I^-(aq)$ Ionic mobility order

4) Which is the correct order of solubility in water :-

(1) $NaCl > SnCl_2 > ZnCl_2$

(2) $ZnCl_2 > SnCl_2 > NaCl$

(3) $SnCl_2 > ZnCl_2 > NaCl$

(4) $SnCl_2 > NaCl > ZnCl_2$

5) Select correct order :-

(1) $LiF < NaF < KF < RbF$ (Lattice energy)

(2) $SiO_2 < GeO_2 < SnO_2 < PbO_2$ (Thermal stability)

(3) $K^+ < Ca^{2+} < Cd^{2+}$ (Polarising power)

(4) $MgF_2 < CaF_2 < BaF_2 < BeF_2$ (Melting point)

6) Which of the following order is incorrect ?

(1) $F^{\ominus}_{(aq.)} < Cl^{\ominus}_{(aq.)} < Br^{\ominus}_{(aq.)} < I^{\ominus}_{(aq.)}$ (conductivity)

(2) AlF_3(Molten) < $AlCl_3$(Molten) (Conductivity)

(3) $Na^+_{(aq.)} < K^+_{(aq.)} < Rb^+_{(aq.)} < Cs^+_{(aq.)}$ (ionic mobility)

(4) ZnS < CdS < HgS (Covalent character)

7) From the following options most effective π-bond is :

(1) 2pπ – 3pπ

(2) 3dπ – 3dπ

(3) 2pπ – 3dπ

(4) 3dπ – 3pπ

8) Which among the following pair does not follows octet rule :

(1) CCl_4, CO_2

(2) IBr, SF_2

(3) BCl_3, PCl_5

(4) NaCl, H_2O

9) Find the incorrect statement :-

(1) Number of hybrid orbitals produced is equal to the number of atomic orbitals that get hybridised.

(2) Hybrid orbitals are equivalent in energy and shape

(3) Hybrid orbitals are more effective in forming sigma bonds than pure atomic orbitals

(4) In hybridisation only half filled atomic orbitals participates

10) The d-orbitals which are involved in hybridisation of central atom in ICl_4^- :-

(1) d_{z^2}, $d_{x^2-y^2}$

(2) $d_{x^2-y^2}$, d_{xy}

(3) d_{z^2}, d_{xy}

(4) d_{xy}, d_{xz}

11) Which of the following is correct :-

(1) $I_3^+ \rightarrow sp^3$ hybridised

(2) $XeOF_4 \rightarrow sp^3d$ hybridised

(3) $SO_2Cl_2 \rightarrow sp^2$ hybridised

(4) $H_3BO_3 \rightarrow dsp^2$ hybridised

12) The shape of XeF_3^+ is :-

(1) Trigonal planar

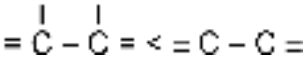
(2) Pyramidal

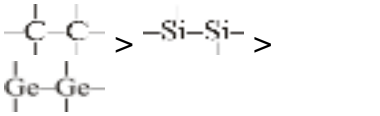
(3) T shape

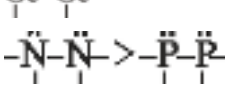
(4) Distorted tetrahedral

13) Which of the following order of BE is not correct ?

(1) C–C > Br–Br > F–F

(2) 

(3) 

(4) 

14) Correct order of bond angle is :-

(1) $OF_2 < H_2O < NH_3 < Cl_2O$

(2) $OF_2 < NH_3 < Cl_2O < H_2O$

(3) $OF_2 < Cl_2O < H_2O < NH_3$

(4) $Cl_2O < OF_2 < H_2O < NH_3$

15) Which of the following pair of compounds are isomorphous.

(1) NaF, CaO

(2) Na_2O, MgS

(3) NaF, MgO

(4) NaCl, MgF_2

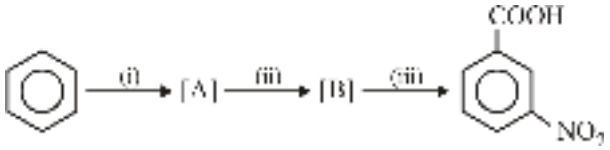
16) Select the incorrect order of stability ?

(1) $Sn^{+2} < Sn^{+4}$

(2) $Sn^{+2} < Pb^{+2}$

(3) $Pb^{+2} < Pb^{+4}$

(4) $Pb^{+4} < Sn^{+4}$

17) 

reagents (i), (ii) & (iii) required for following conversion

(1) (i) H_3C–Cl/Anhydrous $AlCl_3$, (ii) $KMnO_4$, $H^{\oplus}$, (iii) Conc. HNO_3 + Conc. H_2SO_4

(2) (i) Conc. HNO_3 + Conc. H_2SO_4, (ii) $KMnO_4$, $H^{\oplus}$, (iii) H_3C–CH_2Cl/Anhydrous $AlCl_3$

(3) (i) O_3/Zn-H_2O, (ii) $KMnO_4$, $H^{\oplus}$, (iii) Conc. NaOH

(4) (i) CO_2/NaOH, (ii) $CHCl_3$+NaOH, (iii) HNO_2

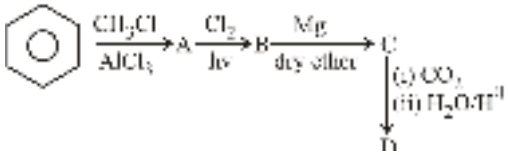
18) **Assertion :** In case of phenol, bromination takes place even in absence of lewis acid whereas bromination of benzene takes place in presence of lewis acid like $FeBr_3$.
Reason : –OH group attached to benzene ring is highly deactivating.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

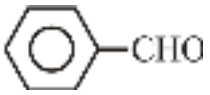
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

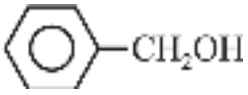
(3) Assertion is True but the Reason is False.

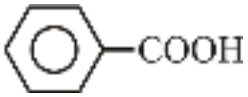
(4) Both Assertion & Reason are False.

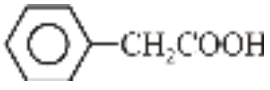
19) 

Product 'D' is :

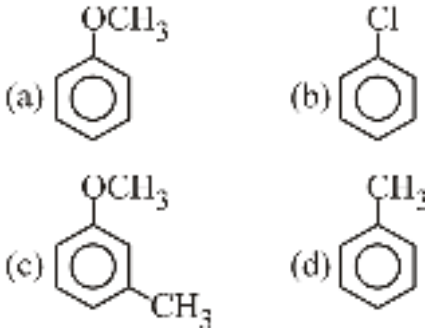
(1) 

(2) 

(3) 

(4) 

20) Correct order of electrophilic substitution reaction is :-



(1) a > b > c > d

(2) b > a > c > d

(3) c > a > d > b

(4) c > d > a > b

21) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution.

Reason (R) : Inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions. In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct

(4) **(A)** is not correct but **(R)** is correct.

22) Friedel craft reaction is not shown in case of -

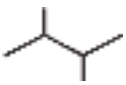
(1) C_6H_6

(2) $C_6H_5OCH_3$

(3) $C_6H_5NO_2$

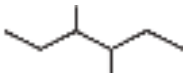
(4) $C_6H_5CH_3$

23) Which of the following is not suitable to obtain from Wurtz reaction :-

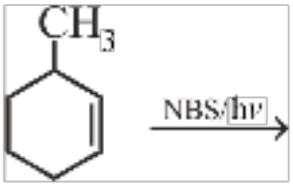
(1) 

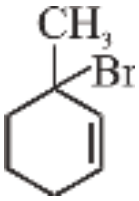
(2) 

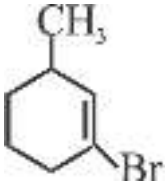
(3) 

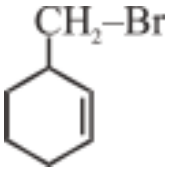
(4) 

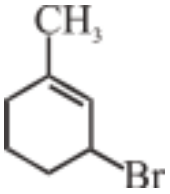
24) The Major Product of following reaction will be :-



(1) 

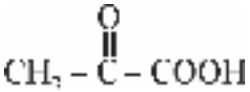
(2) 

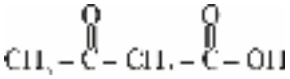
(3) 

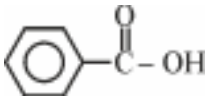
(4) 

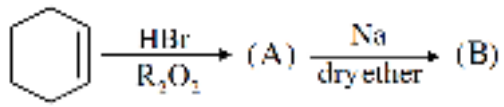
25) Which of the following carboxylic acid undergoes decarboxylation most easily on heating ?

(1) CH_3CH_2COOH

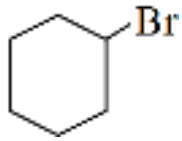
(2) 

(3) 

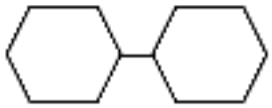
(4) 

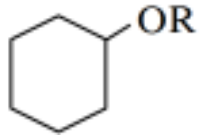
26) 

Product (B) is

(1) 

(2) 

(3) 

(4) 

27) Which of the following will give white crystalline product with $NaHSO_3$:-

(1) Acetophenone

(2) Benzophenone

(3) Acetone

(4) Pentan-3-one

28) Which of the following cannot be obtained as a product by aldol condensation of Acetaldehyde and acetone ?

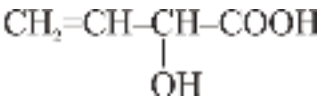
(1) $(CH_3)_2C=CHCOCH_3$

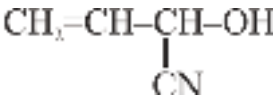
(2) $(CH_3)_2C=CH–CHO$

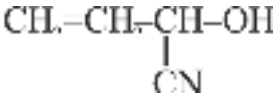
(3) $CH_3CH=C(CH_3)_2$

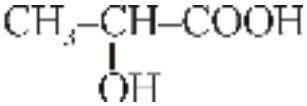
(4) $CH_3CH=CHCHO$

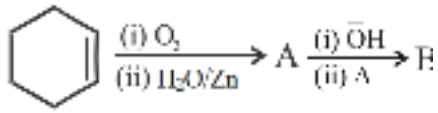
29) $CH_3CHO + HCHO \xrightarrow[\text{heat}]{\text{dil. NaOH}} A \xrightarrow[\text{(ii) } H_3O^{\oplus}]{\text{(i) HCN}} B$
the structure of compound B is :-

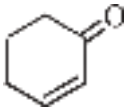
(1) 

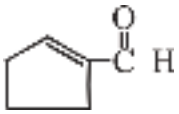
(2) 

(3) 

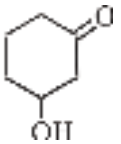
(4) 

30)  ?

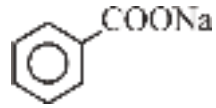
(1) 

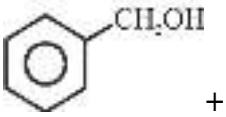
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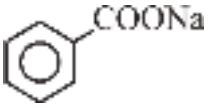
(3) 

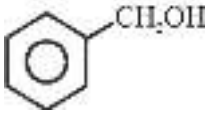
(4) 

31) Reaction of Benzaldehyde with formaldehyde in the presence of 50% conc. NaOH mainly gives :-

(1)  + CH_3OH

(2)  + CH_3CH_2ONa

(3)  +HCOONa

(4) HCOONa + 

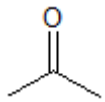
32) In which of the following reaction new carbon-carbon bond is not formed ?

(1) Cannizaro reaction
(2) Wurtz reaction
(3) Friedel-craft Alkylation
(4) Aldol condensation

33)

Which of the following will not give addition reaction with RMgX?

(1) 

(2) 

(3) CO_2
(4) SO_2

34) 2 mole of PCl_5 were heated in a closed vessel of 2 litre capacity. At equilibrium 40% of PCl_5 dissociated into PCl_3 and Cl_2. The value of the equilibrium constant for following reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is :

(1) 0.267
(2) 0.53
(3) 2.63

(4) 5.3

35) The eq^{m} constant for the reaction
$A_4B_2(s) + 5C_2(g) \rightleftharpoons D_2E_4(s)$, is :

(1) $K_C = \frac{1}{[C_2]^5}$

(2) $K_C = [C_2]^5$

(3) $K_C = \frac{[D_2E_4]}{[A_4B_2]}$

(4) $K_C = \frac{[D_2E_4]}{[A_4B_2][C_2]^5}$

36) Value of K_C → 10^{-5} 10^{-2}
Temperature → 400 K 500K
What would you consider by above information :-

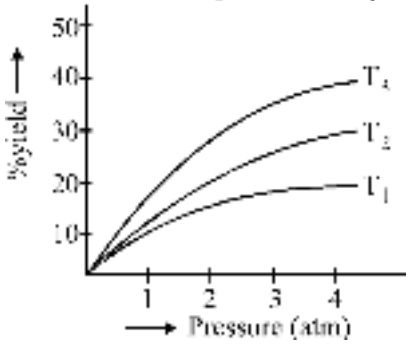
(1) Equilibrium constant increases with increase in concentration.

(2) Heat is absorbed.

(3) Heat is released.

(4) None

37) The preparation of $SO_3(g)$ by reaction $SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$ is an exothermic reaction. If the preparation follows the following temperature-pressure relationship for its % yield, then for



temperatures T_1, T_2 and T_3. The correct options is :-

(1) $T_3 > T_2 > T_1$

(2) $T_1 > T_2 > T_3$

(3) $T_1 = T_2 = T_3$

(4) Nothing could be predicted about temperature through given information

38) For the reaction :-
$H_2O_{(g)} + CO_{(g)} \rightleftharpoons H_{2(g)} + CO_{2(g)}$ K = 0.55,
then the equilibrium constant for
$H_{2(g)} + CO_{2(g)} \rightleftharpoons H_2O_{(g)} + CO_{(g)}$ K_1 = ?

(1) 0.55

(2) $\frac{1}{0.55}$

(3) $(55)^{-1}$

(4) None

39) The equilibrium constant K_p for the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is 1.6 at 200°C. The total pressure at which PCl_5 will be 50% dissociated at 200°C :-

(1) 3.2 atm

(2) 4.8 atm

(3) 2.4 atm

(4) 6.4 atm

40)

The degree of dissociation of PCl_5 (α) obeying the equilibrium, $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, is approximately related to the pressure at equilibrium by (given $\alpha << 1$) :

(1) $\alpha \propto P$

(2) $\alpha \propto \frac{1}{\sqrt{P}}$

(3) $\alpha \propto \frac{1}{P^2}$

(4) $\alpha \propto \frac{1}{P^4}$

41) Ammonium hydrogen sulphide dissociates according to the equation :
$NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$
when solid NH_4HS is taken in an evacuated flask at a certain temperature, It is observed that the pressure of the mixture is 600 mm, when equilibrium is established. Calculated the value of equilibrium constant K_P for the reaction :-

(1) 300 $atm^{2.}$

(2) 760 $atm^{2.}$

(3) 0.156 $atm^{2.}$

(4) None

42) For the gas phase reaction;
$C_2H_{4(g)} + H_{2(g)} \rightleftharpoons C_2H_{6(g)}$;
$\Delta H = -32.7$ kcal carried out in a closed vessel, the number of equilibrium moles of C_2H_4 can be increased by :

(1) Increasing temperature

(2) Decreasing pressure

(3) Removing some amount of H_2

(4) All of these

43) Equilibrium constant of some reactions are given as under :
(a) $x \rightleftharpoons y$ $K = 10^{-1}$
(b) $y \rightleftharpoons z$ $K = 2 \times 10^{-2}$
(c) $P \rightleftharpoons Q$ $K = 3 \times 10^{-4}$
(d) $R \rightleftharpoons S$ $K = 2 \times 10^{-3}$
Initial concentration of the reactants for each reaction was taken to be equal :
Review the above reaction and indicate the reactions in which the reactants and products respectively were of highest concentration :-

(1) d, c
(2) c, a
(3) a, d
(4) b, c

44)

For the reaction $A + B \rightleftharpoons 2C + D$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equillibrium. What should be the concentration of B at equillibrium :-

(1) $\frac{a}{4}$
(2) $\frac{a}{3}$
(3) $\frac{a}{2}$
(4) a

45)

At 527°C, the reaction given below has $K_C = 4$

$$NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$$

What is the K_p for the reaction ?
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(1) $16 \times (800\ R)^2$
(2) $\left(\frac{800R}{4}\right)^{-2}$
(3) $\left(\frac{1}{4 \times 800R}\right)^2$
(4) None of these

BIOLOGY

1) **Statement-I :** When South America join North America, animals of North America are over ridden by animals of south America.

Statement-II : South America has mammals resembling horse, bear, rabbit etc.

(1) Statement-I is correct and statement-II is incorrect

(2) Statement-I is incorrect and statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both statement-I and statement-II are incorrect

2) Which statements are correct regarding continental drift was favourable for -
(a) North America fauna over South American.
(b) South American fauna over North American.
(c) Pouched mammals in Australia over other Mammals.
(d) Other mammals over pouched Mammals in Australia.

(1) a, d

(2) b, d

(3) a, c

(4) b, c

3) Select incorrect option.

(1) The conditions on earth were high temperature volcanic strums, reducing atmosphere containing CH_4, NH_3, O_2, H_2 etc.

(2) In 1953 S.L Miller on Austrelian scientist created condition in a laboratory scale as same as on primetime on earth.

(3) Temperature in miller's experiment was $800^{\circ}C$

(4) Both 1 & 2

4) The evolution can be best explained by :

(1) Theory of inheritance of acquired characters

(2) Theory of descent with modification

(3) Theory of continuity of germplasm

(4) Recapitulation theory

5) Which is not true about fossils ?

(1) Fossils are always the hard parts of past organisms

(2) Mostly fossils are found in sedimentary rocks

(3) Different aged rock sediments contain fossils of different life forms

(4) The age of a fossil can be estimated by radioactive dating

6) Which is correctly matched?

(1) 500 mya – Invertebrates formed

(2) 320 mya – Jawless fishes evolved

(3) 300 mya – First cellular form of life appeared

(4) 65 mya – Dinosaurs appeared

7) Which organisms probably evolved into first amphibians?

(1) Synapsids
(2) Sauropsids
(3) Coelacanth
(4) Ichthyosaurs

8) Identify the correct sequence of periods of mesozoic era:

(1) Devonian → Silurian → Cambrian
(2) Cambrian → Ordovician → Devonian
(3) Triassic → Jurassic → Cretaceous
(4) Triassic → Cretuceous → Jurassic

9) Study the relationship between A and B, and answer D correctly by selecting the correct option.
Thorns of Bougainvillea (A) : Tendrils of Cucurbita (B) :: Bones of forelimbs of Whale (C) : _____(D)_____.

(1) Bones of hindlimbs of Lion
(2) Bones of forelimbs of Snake
(3) Bones of forelimbs of Bat
(4) Bones of hindlimbs of Kangaroo

10) During an archaeological excavation, fossil remains of Stegosaurus were found in layer B of the earth's crust. Another fossil of a different animal was found in layer D, which was one of the layers below the layer B. This new fossil can belong to all of the following, except

(1) Early reptiles
(2) Fishes
(3) Modern birds
(4) Advanced amphibians

11) Match the column I with column II :-

	Column-I		**Column-II**
(i)	Virus	(a)	Adaptive radiation
(ii)	Body hair	(b)	Vestigial
(iii)	Tail in new born baby	(c)	Connecting link
(iv)	Darwin's finches	(d)	Atavism

(1) (i)–(a), (ii)–(b), (iii)–(c), (iv)–(d)
(2) (i)–(c), (ii)–(b), (iii)–(d), (iv)–(a)
(3) (i)–(b), (ii)–(c), (iii)–(a), (iv)–(d)

(4) (i)–(c), (ii)–(b), (iii)–(a), (iv)–(d)

12) Placental mole and marsupial mole shows __A__ and __B__ indicates common ancestor A and B are :-

(1) A→ Divergent Evolution, B→ Analogy
(2) A→ Convergent evolution, B→ Homology
(3) A→ Divergent evolution, B→ Homology
(4) A→ Convergent evolution, B→ Analogy

13) Identify the incorrect match

(1) About 65 MYA - Dinosaurs suddenly disappeared
(2) All vertebrates Heast-Homology
(3) Eye of octopus and mammal - homology
(4) Sweet potato and potato-analogy

14) Golden age of dinasaures was.

(1) Triassic
(2) Jurassic
(3) Devonian
(4) Silurian

15) Which statement supports the embergological evidence of evolution by von baer?

(1) The embryo always pass through the adult stages of other animal
(2) The embryos never pass through the adult stages of other animals.
(3) the embryo occasionally pass through the adult stage of other animal.
(4) All the stages of embryo and adult are similar

16) Which of the following is not consider as a basic factor for evolution.

(1) Over production
(2) Variations
(3) Adaptations
(4) Struggle

17) **Assertion :** Java Ape-man, pecking man and Heidelberg man are the fossils of Homo errectus.
Reason : Homo errectus evolved from homo-habilis.

(1) Both Assertion & Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion & Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

18) Charles Darwin concluded after his sea voyage

(1) Existing living forms share similarities to varying degrees among themselves
(2) Existing living forms share similarities to life forms that existed millions of years ago
(3) Existing living forms are totally dissimilar with life forms that existed millions of years ago
(4) Both (1) and (2)

19) Which statement is wrong for Darwin's theory

(1) All population has built in variation in Characteristics
(2) Organism whose characteristics which are better for survival in natural conditions breed less and survive
(3) Those who are better fit in environment survive more and naturally selected
(4) Natural selection was taken as a mechanism of evolution

20) **Assertion :** Analogous structures are different in appearance with same function.
Reason : Divergent evolution leads to analogy.

(1) Both Assertion & Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion & Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

21) **Assertion :-** Due to continental drift, when south America joined North America, North American mammals were overiden by South American fauna. **Reason :-** Due to the same continental drift, placental mammals of Australia survived because of lack of competition from any other mammals.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but Reason is False.
(4) Both Assertion & Reason are False.

22)

Mark the option with correct matching of column-I with column-II and column-III :-

	Column-I (Specimen)	**Column-II (Geological time)**	**Column-III (Detail)**
(1)	Coelacanth (Lobefins)	350 MYA	The Animal that invaded land first
(2)	Ichthyosaurs	200 MYA	Reptile like fish evolved
(3)	Tyrannosaurs	500 MYA	Evolved from ancestral reptiles

(4)	Triceratops	165 MYA	Extinction due to climatic changes or meteorite collision

(1) 1

(2) 2

(3) 3

(4) 4

23) Which of the following having highest cranial capacity?

(1) Homo habilis

(2) Homo erectus

(3) Homo sapiens neanderthalensis (Neanderthal man)

(4) Homo sapiens fossilis (Cromagnon man)

24) During genetic drift sometime the change in allel frequency is so different in the new population that they become a different species. This effect is called:-

(1) Founder effect

(2) Natural selection

(3) Hardy-Weinberg principle

(4) Bottlenect effect

25)

Read the following statements :-
(A) It is a chronic disorder
(B) Major cause of it is cigarette smoking
(C) It is an Allergy
(D) Alveolar walls are damaged
(E) Respiratory surface is decreased
(F) It is due to inflammation of bronchi and bronchioles.
How many of the above statements are correct about Emphysema disorder :-

(1) Two

(2) Three

(3) Four

(4) One

26) Which of the following situations would result in the greatest degree of O_2 saturation for haemoglobin, assuming PO_2 remains constant-

(1) Increased CO_2 levels, decreased temperature

(2) Increased CO_2 levels, increased temperature

(3) Decreased CO_2 levels, decreased temperature

(4) Decreased CO_2 levels, increased temperature

27)

If total O_2 transported by blood from lungs to cell in a minute is 500 ml. Then the volume of O_2 transported by blood found dissolved in plasma is

(1) 970 ml
(2) 15 ml
(3) 1000 ml
(4) 40 ml

28) Trachea is straight tube which divides at the level of ____ vertebra. Each _____ give rise to a number very thin irregular walled & vascularized bag like structure called alveoli.

(1) 6th Thoracic, & Initial bronchioles
(2) 5th Thoracic & Terminal bronchioles
(3) 5th Cervical & Terminal bronchioles
(4) 5th lumber & Terminal bronchioles

29)

In which of the following gaseous exchange occurs through general body surface?

(1) Sponges
(2) Coelenterates
(3) Flatworms
(4) All of these

30)

Which of the following are involved in the formation of diffusion membrane :
(A) Alveolar epithelium
(B) Basement substance
(C) Capillary Endothelium membrane

(1) C Only
(2) A and B
(3) A and C
(4) A, B and C

31) Match **List-I** with **List-II**:

List-I		**List-II**	
A.	Emphysema	I.	Proliferation of fibrous tissues

B.	Asbestosis	II.	Alveolar walls are damaged and respiratory surface is decreased
C.	Asthma	III.	Serious lung damage
D.	Fibrosis	IV.	Difficulty in breathing due to inflammation of bronchi and bronchioles

Choose the **correct** answer from the options given below :

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-IV, D-III

(4) A-I, B-III, C-IV, D-II

32) In human the process of respiration involves the following steps.
A. Diffusion of gases across alveolar membrane
B. Diffusion of gases between blood and tissues
C. Transport of gases by blood
D. Utilisation of O_2 by the cells for catabolic reactions
E. Breathing or pulmonary ventilation
Choose the correct sequence of steps from the options given below:

(1) C → E → A → B → D

(2) B → D → C → E → A

(3) E → A → C → B → D

(4) A → E → B → C → D

33)

Match List I with List II :

	List I		**List II**
A.	Expiratory capacity	I.	Expiratory reserve volume + Tidal Volume + Inspiratory reserve volume
B.	Functional residual capacity	II.	Tidal volume + Expiratory reserve volume
C.	Vital capacity	III.	Tidal volume + Inspiratory reserve volume
D.	Inspiratory capacity	IV	Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below

(1) A-II, B-IV,C-I,D-III

(2) A-III, B-II,C-IV,D-I

(3) A-II, B-I,C-IV,D-III

(4) A-I, B-III,C-II,D-IV

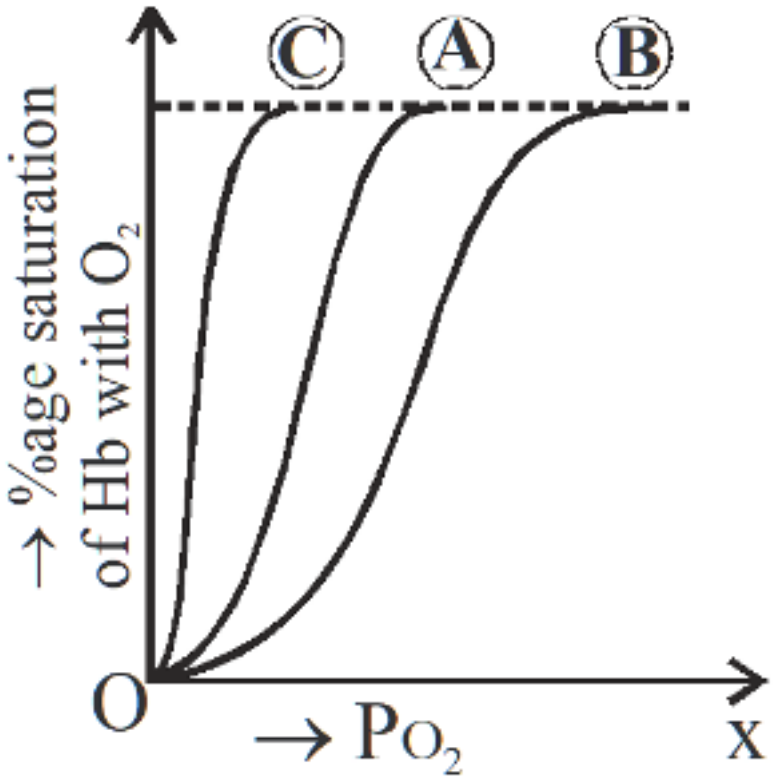
34) The volume of air a person can breath out maximally after forceful inspiration is

(1) ERV + RV
(2) IRV + TV + ERV + RV
(3) IRV + TV + ERV
(4) ERV + TV

35) Which point is incorrect about regulation of respiration ?

(1) Pneumotaxic centre can moderate the functions of the respiratory rhythm centre
(2) Chemosensitive area is highly sensitive to CO2 and hydrogen ions
(3) Respiratory rhythm centre is primarily responsible for respiration regulation
(4) The role of carbon dioxide in the regulation of respiratory rhythm is quite insignificant

36) Given below the graphical representation in which different oxy-haemoglobin dissociation curve shows through A, B and C. Select the correct option which includes all the correct identification of oxyhaemoglobin dissociation curves ?



(ODC- Oxygen-Hb dissociation curve)

(1) A = Foetal ODC
B = Normal ODC
C = Myoglobin ODC

(2) A = Normal ODC
B = Foetal ODC
C = Myoglobin ODC

(3) A = Myoglobin ODC
B = Normal ODC
C = Foetal ODC

A = Foetal ODC
(4) B = Myoglobin ODC
C = Normal ODC

37) From the following relationships between respiratory volumes and capacities, mark the correct option :-
(i) Inspiratory capcity (IC) = Tidal volume
+ Residual volume
(ii) Vital capcity (VC) = Tidal volume (TV)
+ inspiratory Reserve volume (IRV)
+ Expiratory reserve volume (ERV)
(iii) Residual volume (RV) = Vital Capacity (VC)
– Inspiratory reserve volume (IRV)
(iv) Tidal volume (TV) = Inspiratory capacity (IC) – Inspiratory reserve volume (IRV)

(1) (i) incorrect, (ii) Incorrect, (iii) Incorrect, (vi) Correct
(2) (i) Incorrect, (ii) Correct, (iii) Incorrect, (iv) Correct
(3) (i) Incorrect, (ii) Correct, (iii) Incorrect, (iv) Correct
(4) (i) Correct, (ii) Incorrect, (iii) Correct, (iv) Incorrect

38) **Assertion A :** During the transportation of gases about 70 percent of CO_2 is carried by Haemoglobin as carbamino-haemoglobin.
Reason R : Binding of CO_2 with haemoglobin is related to high pCO_2 and low pO_2 in tissues.

(1) **A** is true but **R** is false
(2) **A** is false but **R** is true
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

39) Choose the correct match

A.	Tidal volume (TV)	(I)	1100 to 1200 ml
B.	Inspiratory Reserve volume (IRV)	(II)	1000 to 1100 ml
C.	Expiratory reserve volume (ERV)	(III)	2500 to 3000 ml
D.	Residual volume (RV)	(IV)	500 ml

	A	**B**	**C**	**D**
(1)	I	II	III	IV
(2)	IV	I	II	III
(3)	IV	III	I	II
(4)	IV	III	II	I

(1) 1
(2) 2

(3) 3

(4) 4

40) Match the following columns and select the **correct** option.

	Column - I		***Column - II***
(a)	Bt cotton	(i)	Gene therapy
(b)	Adenosine deaminase deficiency	(ii)	Nematode resistance plant
(c)	RNAi	(iii)	Detection of HIV infection
(d)	*PCR*	(iv)	Insect resistance plant

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(i)	(ii)	(iii)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(ii)	(iii)	(iv)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

41) **Statement A :** The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid of *Salmonella typhimurium*.
Statement B : Transgenic rat, rabbit, pigs, sheep, cows and fish have been produced, although over 95 per cent of all existing transgenic animals are mice.

(1) Both Statement A and B are correct.

(2) Only Statement A is correct.

(3) Only Statement B is correct.

(4) Both Statement A and B are incorrect.

42) Plants, bacteria, fungi & animals whose genes have been altered by manipulation are called genetically modified organisms (GMO). Which of the following statements is not applicable to GM plants ?

(1) Reduced dependence on chemical pesticides

(2) Prevent early exhaustion of fertility of soil

(3) Crops are less tolerant to abiotic stresses (cold, drought, salt, heat)

(4) Enhanced nutritional value of food

43) The proteins encoded by the genes cry I Ac and cry II Ab are effective against :-

(1) Corn borer

(2) Cotton bollworms

(3) Nematodes

(4) Mosquitoes

44) Bt toxin kills insects by-

(1) Obstructing a biosynthesis pathway of sucrose formation

(2) Creating pores in the mid gut epithelial cells leading to cell swelling and lysis.

(3) Generating excessive heats

(4) Inhibiting protein synthesis

45) Given below are some steps of formation of insulin by genetic engineering :-
A. Chain A and B extracted separately.
B. Introduced gene of chain A and B transfer in plasmids of *E. coli*
C. Chain A and B combined by creating disulfide bonds
D. Chain A and B were produced separately
E. Prepared two DNA sequence (Genes) corresponding to A and B chains
Choose the correct sequence of steps from the options given below :-

(1) C-E-D-A-B

(2) E-B-D-A-C

(3) B-D-E-A-C

(4) E-C-A-D-B

46) **Statement-I :** In first gene therapy, functional ADA gene using retroviral vector is introduced into lymphocytes.
Statement-II : The first clinical gene therapy was given to a 4 year old girl with ADA deficiency.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

47) Use of Bioresource by multinational company and other organization without proper authorisation from the countries and people concerned without compensatory payment is known as :

(1) Biopatent

(2) Biowar

(3) Biopiracy

(4) Indian patent bill

48)

The human protein named α-1-antitrypsin, obtained from transgenic animals, is used for the treatment of .....

(1) Alzheimer's disease
(2) Emphysema
(3) Phenyl Ketoneuria
(4) Cystic Fibrosis

49) Which of the following (parts) explants can give rise to a virus-free plant from an infected plant ?

(1) Leaf
(2) Stem
(3) Root
(4) Apical meristem

50) Match the following columns :-

Column-I		**Column-II**	
A.	Ladybird beetle	(i)	Nitrogen fixation
B.	Mycorrhiza	(ii)	Bio gas
C.	Methanogens	(iii)	Aphids
D.	*Rhizobium*	(iv)	*Glomus*

	A	**B**	**C**	**D**
(1)	(ii)	(iv)	(iii)	(i)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(iii)	(ii)	(i)	(iv)

(1) 1
(2) 2
(3) 3
(4) 4

51)

Match the following

(A)	Citric acid	(i)	*Streptococcus*
(B)	Streptokinase	(ii)	*Aspergillus niger*
(C)	Cyclosporin - A	(iii)	*Monascus purpureus*
(D)	Statins	(iv)	*Trichoderma polysporum*

	A	B	C	D
(1)	i	ii	iii	iv
(2)	ii	i	iii	iv
(3)	ii	i	iv	iii
(4)	iv	ii	iii	i

(1) 1

(2) 2

(3) 3

(4) 4

52)

The activated sludge pumped into large tanks. Here, other kinds of bacteria, which grow anaerobically, digest the bacteria and the fungi in the sludge.

This process of sewage treatment performed in which tank ?

(1) Secondary settling tank

(2) Aeration tank

(3) Primary settling tank

(4) Anaerobic sludge digesters

53) Which of the following are fused in somatic hybridization involving two varieties of plants?

(1) Callus

(2) Somatic embryos

(3) Protoplasts

(4) Pollens

54) Main steps in polymerase chain reaction technique (PCR) are given below. Arrange these steps in a correct sequence -
A – Extension
B – Denaturation
C – Annealing

(1) A, B, C

(2) B, C, A

(3) A, C, B

(4) B, A, C

55) Which the following is not a component of plant tissue culture medium ?

(1) Sucrose

(2) Vitamin

(3) Amino acid

(4) Saturated Fatty acid

56) The capacity to generate a whole plant from any cell of the plant is called :

(1) Totipotency
(2) Differentiation
(3) Micropropagation
(4) Somatic hybridization

57) At present, about ……A….. recombinant therapeutics have been approved for human-use in the world over while in India, ……B…….. of these are presently being marketed.
Choose the correct word for A and B blanks :-

(1) A-30, B-12
(2) A-12, B-30
(3) A-31, B-5
(4) A-50, B-5

58) Three observations are made about interaction X:
1. Species A shows increased reproductive success in the presence of B.
2. B also gains a benefit from A.
3. Neither species shows complete dependence on the other for survival.
Which conclusion is most defensible?

(1) X is necessarily obligate mutualism.
(2) X is consistent with a facultative mutualistic/protocooperative association.
(3) X must be parasitism because neither species is dependent.
(4) X is commensalism because both species can survive separately.

59) A plant species releases a compound into soil. After its release, germination of neighbouring plant species decreases, while the releasing plant shows no measurable reduction in growth.
then which interpretation is most appropriate?

(1) A benefits and B is unaffected.
(2) A is harmed and B benefits.
(3) A remains unaffected and B is harmed.
(4) Both A and B are harmed.

60) Species A and B use the same limiting resource. When A is experimentally removed, B increases rapidly. When B is removed, A also increases.
Which inference is strongest?

(1) A and B have a (+,+) interaction.
(2) The interaction between A and B is likely (-,-), even though removal of one benefits the other.
(3) The interaction is (+,0).
(4) The interaction is (-,0).

61) An insect population feeding on a crop increases rapidly. Subsequently, the crop population decreases. After the crop becomes scarce, the insect population also declines.
Which sequence best represents the interaction?

(1) Both populations initially facilitate each other.

(2) Increase in prey/resource availability can support the consumer, while increased consumption can reduce the prey/resource population.

(3) Both populations are independently controlled by competition only.

(4) The crop benefits from the insect, followed by amensalism.

62) A parasite has a much smaller body size than its host, reproduces at a high rate and remains associated with the host for a considerable period. Which combination of features would most strongly support parasitism rather than predation?

(1) Short interaction + immediate killing of host

(2) Prolonged association + nourishment from host + adaptations for attachment

(3) Benefit to both organisms + independent survival

(4) One organism unaffected + one organism harmed

63) Consider the following four situations:
1. Lichen association
2. Cattle egret with grazing cattle
3. Orchid growing on mango tree
4. One plant releasing chemicals inhibiting another plant
Which sequence of interaction types is most appropriate?

(1) Mutualism → Commensalism → Commensalism → Amensalism

(2) Commensalism → Mutualism → Amensalism → Competition

(3) Mutualism → Competition → Commensalism → Predation

(4) Parasitism → Commensalism → Mutualism → Amensalism

64) An organism lives inside another organism, obtains nourishment from it, produces a large number of reproductive units and possesses structures that help it remain attached to the host.
Which combination is most consistent with its ecological relationship?

(1) Commensalism + absence of host dependence

(2) Parasitism + adaptations for successful exploitation of host

(3) Mutualism + equal benefit to host

(4) Amensalism + chemical inhibition of host

65) Which of the following statements is TRUE?

(1) In mutualism, both interacting species are negatively affected.

(2) Lichens represent an association in which both partners derive benefit from each other.

(3) Mutualism always involves two unrelated animal species.

(4) In mutualism, one species benefits while the other remains unaffected.

66) Which of the following statements is TRUE?

(1) Competition occurs only when two species belong to the same genus.
(2) Competition always results in benefit to both interacting species.
(3) Competition may occur when organisms require the same limited resource.
(4) Competition is an interaction in which one species benefits and the other is unaffected.

67) Which of the following statements is TRUE?

(1) Both the species benefit in competition.
(2) Both the species lose in mutualism.
(3) Both the species benefit in mutualism.
(4) One species is harmed whereas the other is unaffected in mutualism.

68) Which of the following statements is TRUE?

(1) In amensalism one species is benefited whereas the other is unaffected.
(2) In commensalism one species is harmed whereas the other is unaffected.
(3) In both parasitism and predation only one species benefits.
(4) In predation both species benefit.

69) Which of the following statements is FALSE?

(1) Predation is nature’s way of transferring to higher trophic levels the energy fixed by plants.
(2) A sparrow eating any seed is no less a predator.
(3) Predation is a very important process through which trophic energy transfer is facilitated.
(4) Predation is an interaction in which both species benefit.

70) Which of the following statements is FALSE?

(1) Interspecific competition is a potent force in organic evolution.
(2) Totally unrelated species could also compete for the same resource.
(3) Resources need not be limiting for competition to occur.
(4) Competition occurs only when closely related species compete for the same resources that are limiting.

71) **Assertion:** Endemic species are highly vulnerable to extinction.
Reason: They are widely distributed across multiple continents.

(1) Both A and R are true, and R explains A
(2) A is true, R is false
(3) A is false, R is true
(4) Both A and R are false

72) **Assertion (A):** Two closely related species cannot coexist in the same niche for long time.
Reason (R): It leads to competitive exclusion.

(1) Both A and R are true, R explains A
(2) Both A and R are true, R does not explain A
(3) A true, R false
(4) Both false

73) Which of the following best defines a niche?

(1) The physical space a species occupies
(2) The habitat where species migrate
(3) The functional role of species in ecosystem
(4) The food chain level of species

74) What does species richness measure?

(1) The total number of individuals
(2) The number of species in a community
(3) The reproductive rate of dominant species
(4) The ecological dominance of one species

75) Which of the following statements about species evenness is incorrect?

(1) It refers to the relative abundance of species
(2) High evenness means species are equally abundant
(3) It increases when a few species dominate
(4) It is used along with richness to assess diversity

76) **Assertion(A):** Two species competing for same resource cannot coexist indefinitely.
Reason (R): One species will eventually outcompete the other (Gause's Principle).

(1) Both A and R are true, and R explains A
(2) A is true, R is false
(3) A is false, R is true
(4) Both A and R are false

77) Which example represents interspecific competition?

(1) Cuscuta and mango
(2) Balanus and Chthamalus on Scottish coast
(3) Koel and Crow
(4) Tiger and deer

78) Which of the following statements correctly explains the ecological concept of stratification in a terrestrial ecosystem?

(1) Division of organisms based on feeding levels
(2) Zonation of organisms based on seasonal migration

(3) Vertical arrangement of different plant species according to their height

(4) Horizontal distribution of energy flow in food chain

79) Match Column I with Column II:

Column I		Column II	
(A)	Natural ecosystem	(1)	Maintained by humans
(B)	Artificial ecosystem	(2)	Self-sustaining to a large extent
(C)	Pond	(3)	Example of a natural ecosystem
(D)	Crop field	(4)	Example of an artificial ecosystem

(1) A–2, B–1, C–3, D–4

(2) A–1, B–2, C–4, D–3

(3) A–2, B–4, C–1, D–3

(4) A–3, B–1, C–2, D–4

80) Which statement best explains the relationship between autotrophs and heterotrophs in an ecosystem?

(1) Autotrophs obtain organic food from heterotrophs, whereas heterotrophs synthesize their own food.

(2) Autotrophs synthesize organic food from inorganic substances, whereas heterotrophs depend on other organisms for food.

(3) Both autotrophs and heterotrophs synthesize their food independently.

(4) Heterotrophs always occupy the first trophic level.

81) Why do autotrophs generally occupy the first trophic level in an ecosystem?

(1) They consume herbivores.

(2) They obtain energy by consuming other organisms.

(3) They convert inorganic resources into organic matter that forms the basic food source for other trophic levels.

(4) They always act as decomposers.

82) Consider the following interaction:
Host → Parasite-X → Parasite-Y
Which statement correctly describes Parasite-Y in this interaction?

(1) It obtains nutrition directly from the host and acts as a primary parasite.

(2) It parasitises Parasite-X and is therefore a hyperparasite.

(3) It is an autotroph because it obtains nutrients from another organism.

(4) It is a decomposer because it depends on a living host.

83) Cuckoo lays its eggs in the nest of a crow, where the crow incubates the eggs and subsequently takes care of the young cuckoo. This interaction is best described as:

(1) Predation, because the cuckoo obtains food from the crow

(2) Social parasitism, because the cuckoo exploits the crow for rearing its young

(3) Mutualism, because both cuckoo and crow benefit from the association

(4) Commensalism, because the crow is neither benefited nor harmed

84) Community X contains 6 species with nearly equal abundance, whereas Community Y contains 10 species, but one species accounts for almost 90% of all individuals. Which statement is correct?

(1) X has lower richness and lower evenness than Y.

(2) X has lower richness but higher evenness than Y.

(3) X has higher richness but lower evenness than Y.

(4) X and Y have equal richness but different evenness.

85) In a forest community, different species of plants occupy different vertical layers. Tall trees form the uppermost layer, while shrubs and herbs occupy progressively lower layers. Which community characteristic is being represented?

(1) Species-area relationship

(2) Stratification

(3) Species richness

(4) Ecological succession

86) In a rocky intertidal experiment, removal of a particular predator resulted in extinction of several invertebrate species within a year. What best explains this result?

(1) Predator was a producer

(2) Predator reduced competition among prey species

(3) Predator directly produced food

(4) Predator converted prey into mutualists

87) Suppose a predator becomes so efficient that it eliminates almost all of its prey. What long-term consequence would be most logical?

(1) Predator population will increase indefinitely

(2) Predator may also decline due to food shortage

(3) Prey will immediately recover

(4) Both will become producers

88) An agricultural pest population rises sharply. Introducing its natural predator reduces the pest population without directly killing every pest. Which ecological principle is being applied?

(1) Predator regulation of prey population

(2) Competitive exclusion

(3) Mutualism

(4) Commensalism

89) Cattle grazing through vegetation disturb insects, making them easier for cattle egrets to catch. The cattle receive no apparent benefit or harm. Which interaction is represented?

(1) Mutualism

(2) Parasitism

(3) Commensalism

(4) Competition

90) A parasitic bird lays its eggs in the nest of another bird. Its eggs closely resemble those of the host. Why would such resemblance be advantageous?

(1) It increases detection by host

(2) It decreases the chance of host rejecting the egg

(3) It prevents egg formation

(4) It converts parasitism into mutualism

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	1	2	3	1	2	1	1	3	1	2	3	3	3	4	2	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	1	2	2	1	2	4	1	2	2	1	1	2	2	4	3	1	4	2	2
Q.	41	42	43	44	45															
A.	1	2	2	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	4	1	1	3	2	3	3	4	1	1	3	4	1	3	3	1	3	4	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	2	1	3	3	3	3	1	2	4	1	1	1	1	2	2	2	2	2
Q.	86	87	88	89	90															
A.	3	4	2	4	3															

BIOLOGY

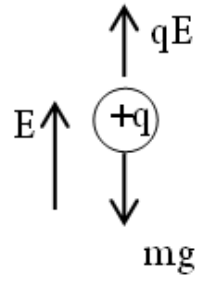
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	4	2	1	1	3	3	3	3	2	2	3	2	2	3	2	4	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	4	1	3	3	2	2	4	4	1	3	1	3	4	1	2	2	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	2	2	2	4	3	2	4	2	3	4	3	2	4	1	1	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	1	2	2	3	3	3	4	4	2	1	3	2	3	1	2	3	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	2	2	2	2	2	1	3	2										

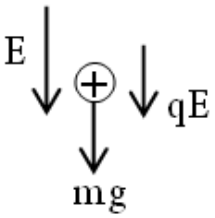
SOLUTIONS

PHYSICS

1)

Let mass of the particle = m, charge on particle = q intensity of electric field between the plates = E initially mg = qE (for equilibrium)





After reversing the field ma = mg + qE
⇒ ma = 2mg
 acceleration of particle
⇒ a = 2g

$$F = -QE = -\frac{Qkqx}{(a^2 + x^2)^{3/2}}$$

2)

q = λ(2πa)

$$F = \frac{-Q(\lambda)(2\pi a)x}{4\pi\in_0(a^2)^{3/2}}, x << a$$

$$F = \frac{-Q\lambda x}{2\in_0 a^2}$$

$$T = 2\pi\sqrt{\frac{m}{\frac{Q\lambda}{2\in_0 a^2}}} = 2\pi\sqrt{\frac{2\in_0 ma^2}{Q\lambda}}$$

3) $\frac{kq^2}{r^2} = \frac{Gm_1m_2}{r^2}$

$$q = \left(\sqrt{\frac{Gm_1m_2}{K}}\right)$$

4) $$E_1 = \frac{kq}{r^2}$$

$$E_2 = \frac{2kq}{r^2} = 2E_1$$

5)

Electric flux $\phi = \vec{E}.\vec{A} = EA\cos\theta$,

For each flat surface $\theta = 0°$

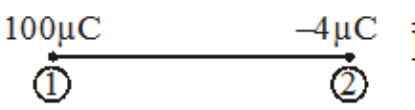
$\phi = EA = E_a(\pi r^2) \times 10^{-4}$ $\frac{Nm^2}{C}$

For curved surface $\vec{E}$ and $\vec{A}$ are perpendicular so
$\phi = 0$

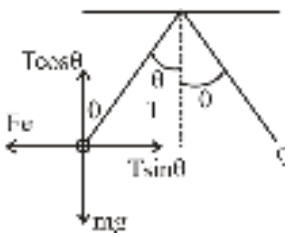
$\therefore \quad \phi = \frac{Q}{\epsilon_0}$

$\Rightarrow Q = \epsilon_0\phi = 8.85 \times 10^{-12} \times 2E_a \times (\pi r^2) \times 10^{-4}$
$= [5.5 \times 10^{-15} E_a r^2]C$

6) The flux passes through ABCD (x – y) plane is zero, because electric field parallel to surface.
Flux of the electric field through surface BCGF (y - z)
At BCGF (electric field) $\Rightarrow \vec{E} = 12\hat{i} - \left(y^2 + 1\right)\hat{j}$
(x = 3m)
Flux $\phi_{II} = 12 \times 4 = 48\ Nm^2/C$
So $\phi_I - \phi_{II} = 0 - 48 = -48\ Nm^2/C$
$\Rightarrow$ Correct answer –48

7) 

$$\vec{F}_{21} = \frac{kq_1q_2}{\left|\vec{r}_{12}\right|^3}\vec{r}_{12}, \vec{r}_{12} = \vec{r}_2 - \vec{r}_1$$

8) 

$T\sin\theta = \frac{Rq^2}{x^2}$ .....(1)
$T\cos\theta = mg$ .....(2)

$$\tan\theta = \frac{Rq^2}{x^2mg}$$

$$\sin\theta = \frac{Rq^2}{x^2mg}$$

$$\frac{x}{2L} = \frac{q^2}{4\pi\varepsilon_0 x^2 mg}$$

$$x = \left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$$

9) Flux coming out of the cube $\phi_1 = \frac{\lambda.a\sqrt{3}}{\varepsilon_0}$ ....(i)

and from sphere $\phi_2 = \frac{\lambda.2a}{\varepsilon_0}$ ...(ii)

$$\frac{\phi_1}{\phi_2} = \frac{\sqrt{3}}{2}$$

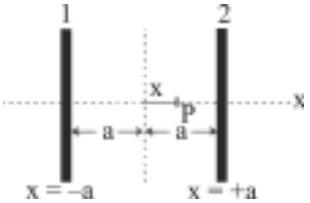
10) At neutral point

$$k \times \frac{20}{(20 \times 10^{-2})^2} = k \times \frac{Q}{(40 \times 10^{-2})^2} \Rightarrow Q = 80\ C$$

11) $\phi = d\frac{q_{in}}{\varepsilon_0} = \frac{\frac{6C}{2} + \frac{8C}{4} + (-7C)}{\varepsilon_0} = \frac{-2C}{\varepsilon_0}$

12) $\oint \vec{E}.\vec{ds} = \frac{q_{in}}{\varepsilon_0}$ $E.4\pi\left(\frac{R}{2}\right)^2 = \frac{\int_0^{R/2} \rho 4\pi r^2 dr}{\varepsilon_0}$

13) Let coordinates of P are (x, 0)



the distance of P from wire 1 = (a + x)

and from wire 2 = (a – x)

$$\because E_1 = E_2 \Rightarrow \frac{2k\lambda_1}{(a + x)} = \frac{2k\lambda_2}{(a - x)}$$

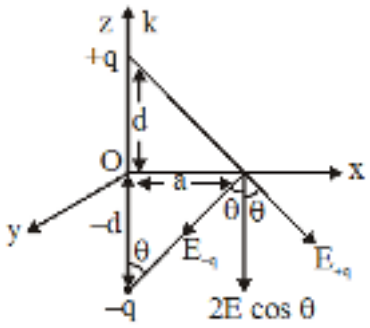
$\Rightarrow \lambda_1(a - x) = \lambda_2(a + x)$

$\Rightarrow \lambda_1 a - \lambda_1 x = \lambda_2 a + \lambda_2 x$

$\Rightarrow (\lambda_1 - \lambda_2)a = (\lambda_1 + \lambda_2)x$

$$\Rightarrow x = \frac{(\lambda_1 - \lambda_2)a}{(\lambda_1 + \lambda_2)}$$

14)



Resultant electric field = 2E cos θ

$$= \frac{1}{4\pi\varepsilon_0}\frac{2q}{(a^2+d^2)}\frac{d}{(a^2+d^2)^{1/2}}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{2qd}{(a^2+d^2)^{3/2}}$$ in –ve Z-direction

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{2qd}{(a^2+d^2)^{3/2}}(-\hat{k})$$

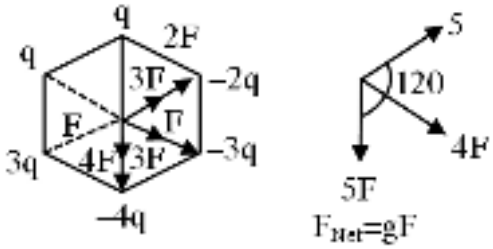
15) $q_{enc} = 10 - 6 = 4\mu C = 4 \times 10^{-6} C$

No. of field lines = $4 \times 10^{-6} - 10^{9}$

$= 4 \times 10^{3}$

So, n = 4

16)



17) $F = qE = \lambda_2 \frac{\lambda_1}{2\pi\in_0 R}$

18) **Explain the question :** In this question, we are asked to find the instantaneous power. We are given with the force vector and velocity vector.

Concept: This question is based on instantaneous power.

Solution: Given here

$\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})N$

$\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k})m/s$

Instantaneous power, $P = \vec{F} \cdot \vec{v}$

$\Rightarrow P = \left(10\hat{i} + 10\hat{j} + 20\hat{k}\right) \cdot \left(5\hat{i} - 3\hat{j} + 6\hat{k}\right)$

P = (10) (5) + (10) (–3) + (20) (6)

P = 50 – 30 + 120 = 140W = 140 joule/sec

Final answer : 3

19) $P = F \cdot v = m\left(v \cdot \frac{dv}{ds}\right) \cdot v$

$\int_{v}^{2v} v^2 \cdot dv = \frac{P}{m}\int_{0}^{s} ds$

$\left[\frac{v^3}{3}\right]_{v}^{2v} = \frac{PS}{m}$

or $S = \frac{7mv^3}{3P}$

20)

P = F·V
= (15000 + 3000)·2
= 36000 W

21)

Asking About :
Power of the engine (actual input power)

Concept :
Efficiency relates useful (output) power to input power :

$$\text{Efficiency} = \frac{\text{Output Power}}{\text{Input Power}} \times 100$$

Formula :
• Output Power = Work done/Time = mgh/t
• Input Power = Output Power / Efficiency

Calculation/Explanation :
Work = mgh = 100 × 10 × 10 = 10000 J

Output Power = $= \frac{10000}{5} = 2000\,W$

Efficiency = 60% ⇒ Input Power = $\frac{2000}{0.6} = 3333.33\,W = 3.33\,kW$

22)

Explain Question : Find instantaneous power delivered to block from a−t graph.
Concept : This question is based on Instantaneous power and kinematics.
Solution :
From the graph (read values) : at t = 4 s the acceleration is a(4) = 2 m/s^2.

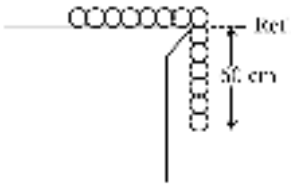
compute change in velocity $\Delta v = \int_{0}^{4} a(t)\,dt$ (area under a-t graph)

Area 0 → 1 (triangle) : $\Delta v_1 = \frac{1}{2} \times 1 \times 2 = 1$ m/s.

Area 1 → 4 (rectangle) : $\Delta v_2 = 3 \times 2 = 6$ m/s.
Total $\Delta v = \Delta v_1 + \Delta v_2 = 1 + 6 = 7$ m/s.
Thus velocity at t = 4 is
v (4) = u + Δv = 2 + 7 = 9 m/s
Instantaneous power
P = ma(4) v(4) = 4 × 2 × 9 = 72 W.
Final answer : 1

23) $\overrightarrow{\Delta p} = \int \vec{F} dt$ = Area under F-t graph
m(v – u) = (30 × 2) – (10 × 1)
10(v – 0) = 50
v = 5 m/s
By work energy theorem,

$$W = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$
$$= \frac{1}{2}(10)(5)^2 - 0$$
$$= 125 \text{ J}$$

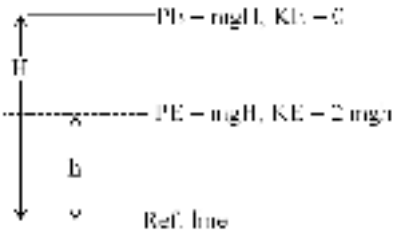
24) 

Mass of hanging chain
$$= \frac{60}{200} \times 4 = \frac{6}{5}\text{kg}$$
So, $\omega_{ent} = \Delta PE$
$$\frac{6}{5} \times 10 \times (\cdot 3)$$
$$= \frac{18}{5} = 3.6\text{J}$$

25)

Apply COME
$$\frac{1}{2}k_1x^2 + \frac{1}{2}k_2x^2 + 0 = \frac{1}{2}mv^2$$
$$(k_1 + k_2)x^2 = mv^2$$

$$\boxed{v = \sqrt[x]{\frac{k_1+k_2}{m}}}$$

26) 

Now, COME
3 mgh = mgh

$$\boxed{h = \frac{H}{3}}$$

Now,

$$KE = 2mg\frac{H}{3} = \frac{1}{2}mv^2$$

$$V = \sqrt{\frac{4gH}{3}}$$

27)

$\omega_{ext} + \omega_g + \omega_f = \Delta KE$
$\omega_{ext} - mgh - \mu_{mg} = 0$
$\omega_{ext} = mgh + \mu_{mg}$
$= mg(h + \mu\ell)$

28)

COME
$K_i + PE_i = K_f + PE_f$
$= 0 + 0 = -2 + KE$

$$\boxed{KE = 2J}$$

29)

$$Cm_1 \rightarrow \frac{\ell}{2}$$

$$Cm_2 \rightarrow \frac{\ell}{2} + \frac{\ell}{2} = \ell$$

$$Cm_3 \rightarrow \frac{\ell}{2} + \frac{\ell}{4} + \frac{\ell}{2} = \frac{5\ell}{4}$$

$$Cm_4 \rightarrow \frac{\ell}{2} + \frac{\ell}{4} + \frac{\ell}{6} + \frac{\ell}{2} = \frac{17\ell}{12}$$

$$x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$$

$$= \frac{m\left(\frac{\ell}{2} + \ell + \frac{5\ell}{4} + \frac{17\ell}{12}\right)}{4m}$$

$$= \frac{1}{4}\left(\frac{6 + 12 + 15 + 17}{12}\right)\ell$$

$$= \frac{25}{24}\ell$$

30) $$x_{cm} = \frac{m_1x_1 - m_2x_2}{m_1 - m_2}$$

In 3rd plates total mass is less than 1st plates.

31) **Question Explanation:**

A. **Given:**
   A. Original disc of radius R and mass M_1 = 9 m centered at (0, 0).
   B. Removed disc of radius $\frac{R}{3}$ centered at $\left(0, \frac{2R}{3}\right)$.
B. **Find:** Position of the center of mass of the remaining disc.

Concept:

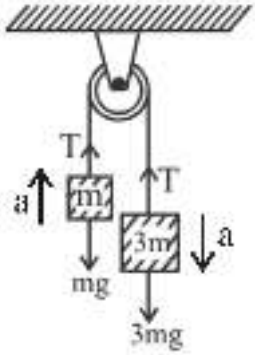
Subtopic: Center of Mass of Remaining Mass ($y_{cm} = \frac{M_1 y_1 - M_2 y_2}{M_1 - M_2}$).
Solution:

1. Find mass of removed disc (M_2) : $M_2 = \frac{\pi(R/3)^2}{\pi R^2} \times 9m = \frac{1}{9} \times 9m = m$ 2. Calculate Y_{cm} of remaining part: $X_{cm} = 0$ $Y_{cm} = \frac{(9m)(0) - (m)\left(\frac{2R}{3}\right)}{9m - m} = \frac{-\frac{2R}{3}}{8} = -\frac{R}{12}$ Position $= \left(0, -\frac{R}{12}\right)$ **Final Answer:** Option (1)

32) **Question Explanation :** The system consists of two masses, m_1 = m and m_2 = 3m, connected by a string over a pulley. The acceleration of the center of mass a_{CM} for the system

Concept: Acceleration of Center of Mass.

Solution:



Let the system moves with an acceleration a as shown in the figure.

3 mg – T = 3 ma

T – mg = ma

2mg = 4ma or a = g/2

Now the acceleration of the center of mass,

$\vec{a}_{cm} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2} = \frac{3m\left(-\frac{g}{2}\right) + m\left(\frac{g}{2}\right)}{3m + m} = -\frac{g}{4}$

Final answer: option (1)

33) **Explain question:**
The question asks about distance travelled by plank when A and B exchange their position on plank.

Concept:
In the absence of external forces, the COM of a system remains stationary.

Solution :
Velocity of COM remains same in the absence of external forces.
So COM of system remains stationary. In this situation displacement of plank w.r.t. ground,
$\Delta x = \frac{90(12) + 40(-12)}{90 + 40 + 20} = 4$
Plank moves by 4m when when A and B exchange their positions.

Final Answer: Option (2)

34) **Explaining :** Find velocity of third fragment.
Concept : This question is based on conservation of linear momentum.
Solution : $3m(200 \cos 60°)\hat{i} = (m \times 100)\hat{j} + (m \times 100)\left(-\hat{j}\right) + m\vec{v}$ $\vec{v} = \left(300\,\hat{i}\right)$ m/s
$\vec{v}$ = 300 m/s in the horizontal direction.
Final Answer : (2)

35) $0 = 1 \times 60\hat{i} + 1 \times 60\hat{j} + 3\nu'$
$\nu' = \frac{-60\hat{i} - 60\hat{j}}{3}$
$v' = -20\hat{i} - 20\hat{j}$
$\left|\vec{\nu}'\right| = 20\sqrt{2}$ m/s

36) **Explain Question:**
A bullet enters a stationery piece of ice the lying friction less plane than find velocity

Concept:
Conservation of momentum.

Solution:
By using C.O.L.M
$m_1\vec{v}_1 + m_2\vec{v}_2 = (m_1 + m_2)\overrightarrow{V_{com}}$
$\left(10 \times 10^{-3}\right)(500) + 0 = \left(\left(10 \times 10^{-3}\right) + 10\right)V$
$V = \frac{5}{10.01} = 0.5$m/s

Final Answer: Option (3)

37)

From COM
$mV_1 + M(0) = (M + m)v$
$v = \frac{mv_1}{M + m}$
From COE
$(M + m)gh = \frac{1}{2}(M + m)v^2$
$2gh = \frac{m^2v_1^2}{(M + m)^2}$

$V_1 = \frac{M+m}{m}\sqrt{2gh}$

38)

$F = 0 \Rightarrow 1 - \frac{t}{T} = 0 \Rightarrow t = T$

Now,

$\frac{dv}{dt} = F_0\left(1 - \frac{t}{T}\right)$

$\int_0^v dv = \int_0^T f_0\left(1 - \frac{t}{T}\right) dt$

$v = f_0\left[t - \frac{t^2}{2T}\right]_0^T$

$v = f_0\left[T - \frac{T}{2}\right] = \frac{f_0 T}{2}$

39)

$H = \frac{1}{2} gt^2$

$H - 60 = \frac{1}{2} g(t-2)^2$

$\frac{1}{2} g[t^2 - (t-2)^2] = 60$

$\frac{1}{2}[(2t-2)(2)] = 6$

$2t - 2 = 6$

$t = 4$ sec

$H = \frac{1}{2} \times 10 \times 16 = 80$ m

40) Velocity becomes zero after 6 s.

$0 = u - g \times 6 \Rightarrow u = 60$ m/s

Distance in the first second

$s_1 = ut - \frac{1}{2} gt^2 = 60 \times 1 - \frac{1}{2} \times 10 \times 1^2 = 55$ m

After 6 s, particle is moving in downward direction.

Distance in the 7th second, $s_2 = \frac{1}{2} \times 10 \cdot 1^2 = 5$ m

$\frac{s_1}{s_2} = \frac{55}{5} = \frac{11}{1}$

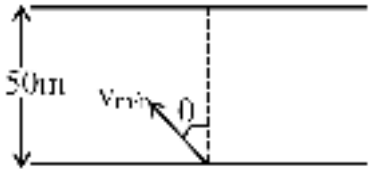
41)

$v_1 + v_2 = 6$ .....(1)

$v_1 - v_2 = 4$ .....(2)
from equation (1) & (2) : $v_1 = 5$m/s & $v_2 = 1$ m/s

42)

$$S_{rel} = u_{rel} \times t + \frac{1}{2}a_{rel} \times t^2$$
$$1.5 = \frac{1}{2}(10+2) \times t^2$$
$$t^2 = \frac{1}{4} \Rightarrow t = \frac{1}{2}\text{ sec}$$

43) 

$$\Rightarrow t_{min} = \frac{50}{v_{m/r}} = 10$$
$v_{m/r} = 5$ m/min
⇒ Now for shortest path
$v_{m/r} \sin\theta° = 3$
$$\sin\theta^\circ = \frac{3}{5} \Rightarrow 37^\circ$$
So, angle with river flow
= 90° + 37° = 127°

44)

$x = 3t^2 - 6t$ $y = t^2 - 2t$
$v_x = \frac{dx}{dt} = 6t - 6$ $v_y = \frac{dy}{dt} = 2t - 2$
at t = 1 sec → $v_x = 0$ and $v_y = 0$
hence $v = \sqrt{v_x^2 + v_y^2} = 0$

45) $v_x = \frac{dx}{dt} = 3\alpha t^2$ and $v_y = \frac{dy}{dt} = 3\beta t^2$
So, speed = $\sqrt{v_x^2 + v_y^2} = 3t^2\sqrt{\alpha^2 + \beta^2}$

CHEMISTRY

46) **(I) NH_3 [1 lone pair]**
(II) SF_4 [1 lone pair]
(IV) XeF_6 [1 lone pair]

47)

Lattice energy ∝ Product of charge of cation and anion.

48)

Hydration ↑ Hydrated size ↑ Ionic mobility ↓

49)

Solubility in water $\propto \frac{1}{\text{Polarisation}}$

50)

Polarising power $K^{+} < Ca^{+2} < Cd^{+2}$

51)

AlF_3 is ionic
$AlCl_3$ is covalent

52)

B.S. $\propto \frac{1}{\text{no. of shell}}$
B.S. ∝ Directional Nature

53) **BCl_3 (Boron Trichloride):**
• Boron has only 6 electrons around it in BCl_3 (3 bonds), making it electron-deficient or hypovalent.

PCl_5 (Phosphorus Pentachloride):
• Phosphorus has 10 electrons around it in PCl_5 (5 bonds), exceeding the octet rule, making it hypervalent.

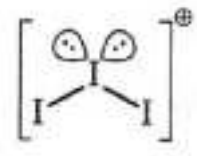
54)

Half filled, full filled as well as vacant orbitals can participate in overlapping.

55)

The ICl_4^- ion has sp^3d^2 hybridization. For this geometry, the two d-orbitals involved are dz^2 and dx^2-y^2

56) $I_3^+ \Rightarrow 2\sigma + 2\ell p = sp^3$ Hyb.



57) With 2 lone pairs in trigonal bipyramidal arrangement → **T-shaped geometry**.

58) B.E. $\propto$ % S character $\propto \frac{1}{\text{size}}$ $\propto \frac{1}{\text{no. of L.P.}}$

59) $NH_3 \Rightarrow 107°$, $OCl_2 \Rightarrow > 109.5°$
$H_2O \Rightarrow 104.5°$, $OF_2 \Rightarrow 103°$

60)

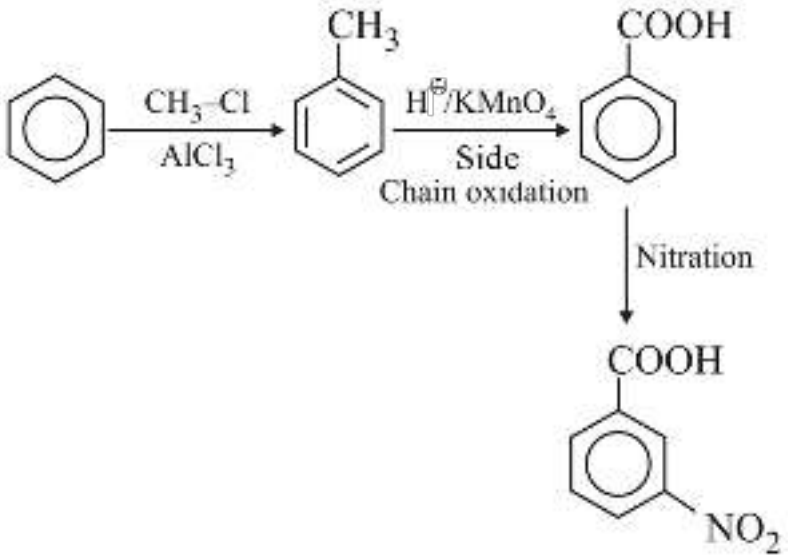
Number of ions and electronic structure are same so, NaF, MgO are isomorphous.

61)

$Pb^{2+} > Pb^{4+}$ (due to inert pair effect)

62)

Concept : This question is based on Friedel craft reaction, side chain oxidation, nitration (ESR).
Solution :



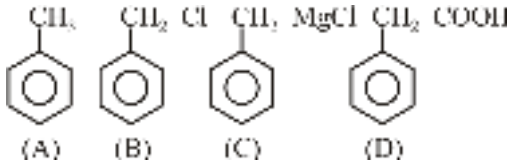
Benzoic acid is meta directing.
Final answer : Option (1)

63)

Explain Question: Evaluate the truth and logical connection between the assertion and reason regarding bromination of phenol vs benzene.
Concept : Electrophilic Aromatic Substitution, Activating/Deactivating Group Effects.
Solution:

A. **Assertion:** True–phenol undergoes bromination **without Lewis acid** due to the activating effect of –OH group.
B. **Reason:** False– –OH is **strongly activating**, not deactivating. It increases electron density on the ring via resonance, especially at ortho and para positions.
C. Therefore, the **reason contradicts** the actual behavior of –OH group.

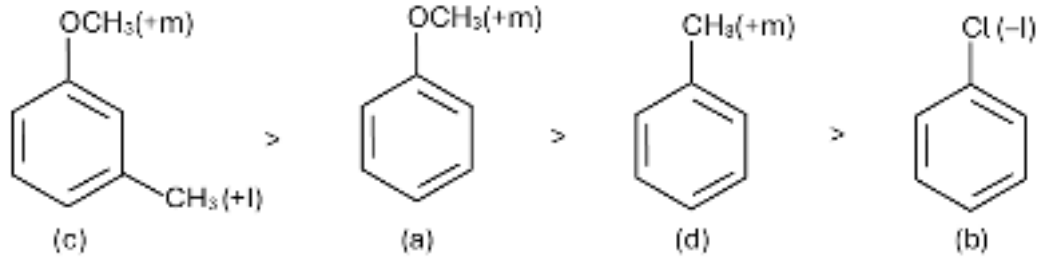
Final Answer: Option (3)

64) 

65)

Concept :

This question is based on Rate of ESR $\propto \frac{e^{\ominus} \text{ donating group}}{e^{\ominus} \text{ withdrawing group}}$
Solution :



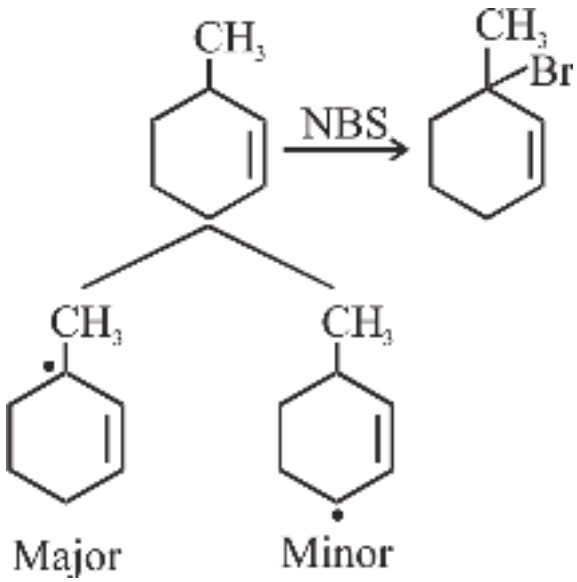
Hence,
Option (3) is correct

66) Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution because Inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions.

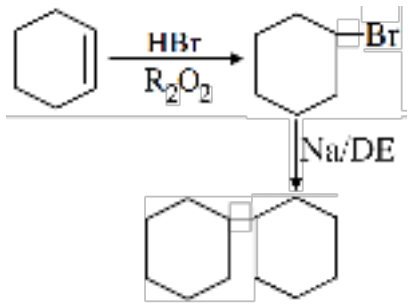
67)

Nitrobenzene has $–NO_2$ which has very strong –M and it strongly deactivating group. So, $Ph-NO_2$ does not give friedel craft reaction.
Final Answer : Option (3)

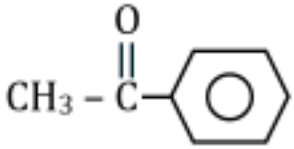
68) Wurtz reaction not suitable to prepare unsymmetrical alkane.

69) 

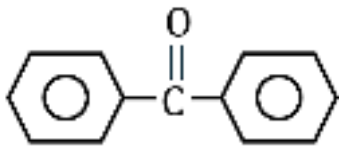
70) β-keto acids are easily decarboxylated

71) 

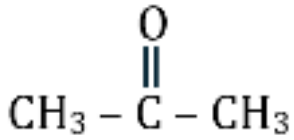
72) **Concept:** NAR of carbonyl compound.
Solution: The reaction of carbonyl compounds with $NaHSO_3$ is steric sensitive since attacking nucleophile is bulky so only following carbonyl compounds give white ppt with $NaHSO_3$.
(A) All aldehydes
(B) Aliphatic methyl Ketone.
So we will check all options.
(1) Acetophenone



Not give reaction
(2) Benzophenone

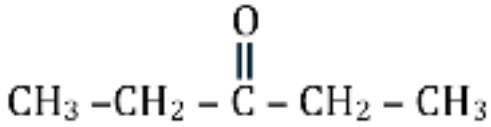


Not give reaction
(3) Acetone



Give white ppt.

(4) Pentan-3-one



Not give reaction

Final Answer: Option (3)

73)

Concept: Aldol Condensation, Enolate Formation, Dehydration, α H

Solution:

Product	Possible Source	Can form?
$(CH_3)_2C=CHCOCH_3$	Self aldol of acetone	Yes
$(CH_3)_2C=CH-CHO$	Cross aldol: acetaldehyde enolate + acetone	Yes
$CH_3CH=C(CH_3)_2$	No carbonyl group in product	No
$CH_3CH=CHCHO$	Self aldol of acetaldehyde	Yes

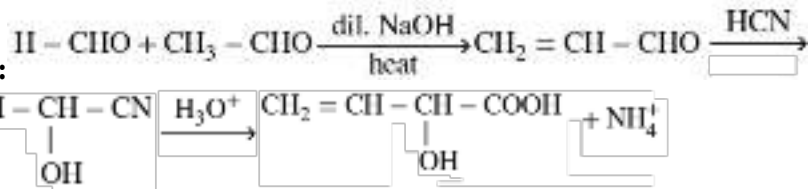
Aldol condensation products retain a **carbonyl group** after dehydration. Option (3) has **no carbonyl group**, so it cannot be formed.

Final Answer: Option (3)

74)

Concept: Aldol Reaction, Dehydration, HCN Addition, Nitrile Hydrolysis

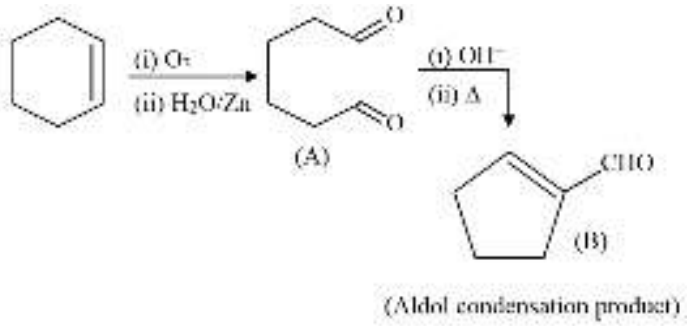
Solution:



Final Answer: Option (1)

75) **Concept:** Ozonolysis and aldol condensation.

Solution:

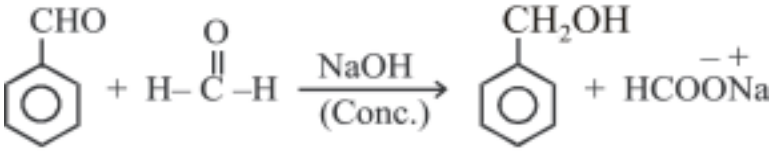


(Aldol condensation product)

(A) is obtained by ozonolysis of cyclohexene.
(B) is obtained by intramolecular aldol condensation of (A).
Final Answer: Option (2)

76) **Concept :** Crossed Cannizzaro reaction, nucleophilic attack, organic oxidation-reduction
Solution :
Both benzaldehyde (C_6H_5CHO) and formaldehyde (HCHO) lack α-hydrogen atoms. When treated with concentrated alkali (50% NaOH), they undergo a **Crossed Cannizzaro reaction**. Since formaldehyde (HCHO) is sterically less hindered and has a more electrophilic carbonyl carbon than benzaldehyde, nucleophilic attack by OH^- occurs preferentially on formaldehyde.

A. Formaldehyde is selectively **oxidized** to form sodium formate (**HCOONa**).
B. Benzaldehyde undergoes reduction by accepting a hydride ion (H^-) to form benzyl alcohol (**$C_6H_5CH_2OH$**).



(Cannizaro Reaction)

Therefore, the main products formed are sodium format (HCOONa) and benzyl alcohol ($C_6H_5CH_2OH$), which is **Correct**.
Final Answer: Option (4).

77)

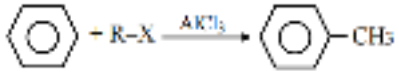
Concept:- Cannizzaro reaction, Aldol condensation, wurtz reaction, FCR Alkylation.
Solution :-
(1) Cannizzaro Reaction:-

$$HCHO + HCHO \xrightarrow[\Delta]{\text{Conc. NaOH}} CH_3OH + HCOONa$$

This reaction involves the transfer of hydride ion and Redox, but no new carbon-carbon bond are formed in product.
(2) Wurtz Reaction:-

$$2R-x \xrightarrow{2Na} R-R+2Nax$$

Two Alkyl group combined, forming a new C-C bond.
(3) Friedel-cratts reaction.

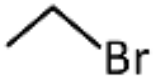


Alkyl group form a new C-C bond with aromatic ring.
(4) Aldol condensation.

$$2CH_3CHO \xrightarrow[\Delta]{NaOH} CH_3-CH=CH-CHO$$

New carbon-carbon bond formed in product.
So in cannizzaro reaction new C-C bond is not formed.
Final Answer :- Option (1)

78) **Concept :**



doesn't give addition reaction with Grignard Reagent
Ans option (1)



79) **Explain the question :** We need to find equilibrium constant if %a and initial moles of PCl_5 with container volume are given.
Concept : This question is based on degree of dissociation with K_C.
Solution :

	$PCl_5 \rightleftharpoons PCl_3 + Cl_2$		
	(g)	(g)	(g)
t = 0	2	–	–
t_{eq}	2–x	x	x

$$\%\alpha = 40\% ; \frac{40}{100} = \frac{x}{2} \Rightarrow x = 0.8$$

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{0.8}{2} \times \frac{0.8}{2} \times \frac{2}{1.2}$$

$K_C = 0.267$
Final answer : 1

80) $$K_c = \frac{1}{[C_2]^5}$$

81) **Explain Question:**
To select correct option from the given data of temp and k_{eq}.

Concept:
Effect of temp on k_{eq}.

Solution:
kc 10^{-5} 10^{-2}
Temp. 400k 500k
On increasing temp. kc increases so reaction is ectothermic

Final answer: Option (2)

82)

Explanation Question:
The question asks to determine the relationship between temperatures T_1, T_2, and T_3 based on the graph for an exothermic reaction.

Concept:
This question is based on Le Chatelier's Principle
Solution:
For an exothermic reaction, a lower temperature favors a higher yield of the product.
The graph shows that at a constant pressure, the yield increases as we go from the curve for T_1 to T_2 to T_3.
The yield relationship is: $T_3 > T_2 > T_1$. Since this is an exothermic reaction, a higher yield means a lower temperature. Therefore, the temperature relationship must be the inverse of they yield relationship $T_1 > T_2 > T_3$
Final Answer: (1)

83) $K_C^1 = \frac{1}{K_C}$

84) $k_P = \frac{\alpha^2}{1-\alpha^2} \times P_T$
$1.6 = \frac{0.25}{0.75} \times P_T$
$P_T = 4.8$

85) **Question Explanation :**
The question asks to find the approximate relationship between the degree of dissociation (α) and the total equilibrium pressure (P).
Concept : This question is based on relationship between α, K_p, and P.
Solution :

	$PCl_5(g) \rightarrow$	$PCl_3(g)$ +	$Cl_2(g)$
moles at eq	1–α	α	α

$K_P = \frac{\left(P_{PCl_3}\right)\left(P_{Cl_2}\right)}{P_{PCl_5}} = \frac{\left(\frac{\alpha P}{1+\alpha}\right)\left(\frac{\alpha P}{1+\alpha}\right)}{\left(\frac{1-\alpha}{1+\alpha}\right)P} = \frac{\alpha^2 P}{1-\alpha^2}$ Since K_p, is constant, $\alpha^2 \propto \frac{1}{P}$, which leads to $\alpha \propto \frac{1}{\sqrt{P}}$
Final Answer : option (2)

86)

$K_P = P_{NH_3} \times P_{H_2S}$
$= \frac{300}{760} \times \frac{300}{760} = 0.156$

87)

Question Explanation
We need to find which change will increase the equilibrium moles of C_2H_4 in the exothermic reaction $C_2H_4 + H_2 \rightleftharpoons C_2H_6$, where $\Delta H = -32.7\text{kcal}$.

Concept: This question is based on (Le Chatelier's principle)

Solution
Reaction: $C_2H_4(g) + H_2(g) \rightleftharpoons C_2H_6(g) \Delta H = -32.7kcal$ (exothermic)
• Increasing temperature → shifts equilibrium backward (more C_2H_4)
• Decreasing pressure → increases total moles (shifts backward)
• Removing H_2 → shifts backward to form more reactants

All these actions shift equilibrium backward, increasing C_2H_4 moles.

Final Answer
Option (4)

88)

Based on fact

89)

	A	+	B ⇋	2C	+	D
t = 0	a		1.5a			
Equation	(a – x)		(1.5a – x)	2x		x

(A) = (D)
a – x = x
x = a/2
(B) = a

90) **Explanation:**
Reaction: $NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$ $K_C = 4$
Reverse reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2\ NH_3(g)$ K_C'
Concept: Relation between K_p and K_p
Solution:
The relation between Kp and Kc is given by:

$K_C' = \frac{1}{(K_C)^2} = \frac{1}{(4)^2}$

$K_P = K_C(RT)^{\Delta n_g}$

$= \frac{1}{(4)^2}(RT)^{-2}$

$K_P = \left(\frac{1}{4 \times 800\ R}\right)^2$

Answer is option (3).

BIOLOGY

91) NCERT Pg. No. # 124

92) NCERT Pg. No. # 124

93) NCERT Pg. No. # 111, 112

94)

NCERT, Pg. # 124

95)

NCERT, Pg. # 124

96)

NCERT, Pg. # 133

97)

NCERT, Pg. # 134

98)

NCERT, Pg. # 134

99)

NCERT, Pg. # 126

100)

NCERT, Pg. # 124

101) Module

102) Module

103) Module

104) NCERT, Pg. # 125

105) NCERT Page No. 113

106) NCERT, Pg. # 124

107)

NCERT, Pg. # 136

108)

NCERT Pg # 128/129

109)

NCERT Pg # 128-129

110)

NCERT, Pg. # 128

111) NCERT Pg#140, para-7.8

112)

NCERT Pg. # 134

113)

NCERT Pg. # 137

114)

NCERT Pg. # 132

118) **Explain the Question :** Fill in the blanks related to trachea and alveoli formation.
Concept : This question is based on human respiratory system – trachea
Solution :
The trachea is a straight tube that divides at the level of the 5th thoracic vertebra (T_5) into right and left primary bronchi.
Terminal bronchioles further give rise (through respiratory bronchioles and alveolar ducts) to thin, irregular-walled, highly vascularized alveoli, where gas exchange occurs.
This matches Option 2.
Final Answer : (2)

119)

NCERT-XIth, Pg.# 183

120)

NCERT-XIth, Pg.# 188

121) NCERT-XI, Pg. # 190, 191

122) NCERT-XI, Pg. # 185

123) NCERT-XI, Pg. # 186, 187

124)

NCERT-XIth, Pg.# 187

125)

NCERT XI, Pg # 190

126)

Module-5

127)

NCERT-XIth, Pg.# 187

128)

NCERT-XIth, Pg.# 189

129)

NCERT-XIth, Pg.# 187

130) **(a)** Bt cotton → (iv) Bacillus thuringiensis
(b) Adenosine deaminase deficiency → (i) Gene therapy
(c) RNAi → (ii) Cellular defence
(d) PCR → (iii) Detection of HIV infection

131) NCERT Pg. # 164, 183

132) **Explain Question:**
The question asks which statement is not true (i.e., not applicable) to genetically modified (GM) plants.

Concept :
Genetically modified (GM) plants. (BIOTECH. APPLICATION IN AGRICULTURE)

Explanation :
Option 3 is incorrect because GM crops are usually more tolerant to abiotic stresses, not less. Hence, this statement is not applicable to GM plants.

Final answer: 3

133)

NCERT (XIIth) Pg. # 209

134) NCERT - 208 (12.1), 209

135) NCERT Pg. # 182

136)

Question Explanation: The question asks about the first clinical gene therapy and its details.

Concept: The question is based on gene therapy for ADA deficiency.

Solution: Both statements are true and correctly describe the first clinical gene therapy.

A. **Statement I**: Correct. In the first gene therapy, a functional ADA gene was introduced into lymphocytes using a retroviral vector.
B. **Statement II**: Correct. The first clinical gene therapy was performed in 1990 on a 4-year-old girl suffering from ADA deficiency.

Final Answer: Option 4

137) **Biopiracy** is the unauthorized use of biological resources by multinational companies or organizations without proper authorization or compensatory payment to the countries or people concerned.
Hence, the correct answer is option (3).

138)

NCERT XII pg. # 184

139) NCERT XII, Pg. # 177

140) NCERT-XII, Pg. # 156,157,158

141)

NCERT-XII, Pg. # 183

142) NCERT XII, Pg. # 184, Para-10.3

143) NCERT 12th (Eng.) Page No :- 178,179

144) NCERT Pg.# 172

145)

NCERT XII Pg # 178

146) NCERT 12th (Eng.) Page No :-178,179

147) NCERT XII Pg # 181,182,183

148) Benefit to both identifies the mutualistic direction; absence of obligatory dependence changes the nature of the association, not the sign.

149) Chemical inhibition can produce a (-,0) relationship.

150) Observe the interaction while both species coexist. The increase after removal is evidence of release from competition.

151) Predation produces (+,-); predator population is dependent on prey availability.

152) Parasites are often host-specific and show adaptations such as loss of unnecessary sense organs, presence of adhesive organs/suckers, and high reproductive capacity.

153) Identify the effect on each partner rather than identifying the habitat relationship alone.

154) Host exploitation together with attachment and reproductive adaptations is characteristic of parasitic association.

155)

Conceptual Understanding → Ecological Interactions

156) Conceptual Understanding → Resource Limitation → Competition

157)

Module

158)

Module

159)

Module

160) *Organisms and Populations — Population Interactions*

161)

NCERT-XIth, Pg.# 255

162)

NCERT-XIth, Pg.# 246

163)

NCERT-XIth, Pg.# 246

164)

NCERT-XIth, Pg.# 257

165)

NCERT-XIth, Pg.# 257

166)

NCERT-XIth, Pg.# 246

167)

NCERT-XIth, Pg.# 246

168)

NCERT-XIth, Pg.# 248

169)

Module

170)

Module

171)

They convert inorganic resources into organic matter that forms the basic food source for other trophic levels.

172)

Module

173)

Module

174)

Module

175)

Module

176) NCERT-XIth, Pg.# 198

177) NCERT Pg. No. # 198

178) NCERT Pg. No. # 198

179) NCERT Pg. No. # 201

180) NCERT Pg. No. # 201